CHAPTER
3

Nervous system

The nervous system has two major divisions, the central nervous system (CNS) and the peripheral nervous system (PNS). The CNS consists of the brain, spinal cord, optic nerve and retina, and contains the majority of neuronal cell bodies. The PNS includes all nervous tissue outside the CNS and consists of the cranial and spinal nerves, the peripheral autonomic nervous system (ANS) and the special senses (taste, olfaction, vision, hearing and balance). It is composed mainly of the axons of sensory and motor neurones that pass between the CNS and the body. The ANS is subdivided into sympathetic and parasympathetic components. It consists of neurones that innervate secretory glands and cardiac and smooth muscle, and is concerned primarily with control of the internal environment. Neurones in the wall of the gastrointestinal tract form the enteric nervous system (ENS) and are capable of sustaining local reflex activity that is independent of the CNS. The ENS contains as many intrinsic neurones in its ganglia as the entire spinal cord and is often considered as a third division of the nervous system (Gershon 1998).

In the CNS, the cell bodies of neurones are often grouped together in discrete areas termed nuclei, or they may form more extensive layers or masses of cells; collectively they constitute the grey matter. Neuronal dendrites and synaptic contacts are mostly confined to areas of grey matter and form part of its meshwork of neuronal and glial processes, termed the neuropil. Their axons join bundles of nerve fibres that tend to be grouped separately to form tracts. In the spinal cord, cerebellum, cerebral cortices and some other areas, concentrations of tracts constitute the white matter, so called because the axons are often ensheathed in lipid-rich sheaths of myelin, which is white when fresh (**Fig. 3.1**; see Fig. 16.9).

The PNS is composed of the efferent axons (fibres) of motor neurones situated inside the CNS, and the cell bodies of sensory neurones (grouped together as ganglia) and their afferent processes. Sensory cells in dorsal root ganglia give off both centrally and peripherally directed processes; there are no synapses on their cell bodies. Also included are ganglionic neurones of the ANS, which receive synaptic contacts from the peripheral fibres of preganglionic autonomic neurones whose cell bodies lie within the CNS. For further details of the organization of the nervous system, see Chapter 16.

When the neural tube is formed during prenatal development (Sanes et al 2011), its walls thicken greatly but do not completely obliterate the cavity within. The latter remains in the spinal cord as the narrow central canal and becomes greatly expanded in the brain to form a series of interconnected cavities called the ventricular system. In the fore- and hindbrains, parts of the neural tube roof do not generate neurones but become thin, folded sheets of secretory tissue, which are invaded by blood vessels and are called the choroid plexuses. The latter secrete cerebrospinal fluid (CSF), which fills the ventricles and subarachnoid spaces, and penetrates the intercellular spaces of the brain and spinal cord to create their interstitial fluid. The CNS has a rich blood supply, which is essential to sustain its high metabolic rate. The blood–brain barrier places considerable restrictions on the substances that are able to diffuse from the blood stream into the neuropil.

Neurones encode information, conduct it over considerable distances, and then transmit it to other neurones or to various non-neural targets such as muscle cells. The propagation of this information within the nervous system depends on rapid electrical signals, the action potentials. Transmission to other cells is mediated by secretion of neurotransmitters at special junctions, either with other neurones (synapses), or with cells outside the nervous system, e.g. muscle cells at neuromuscular junctions, gland cells or adipose tissue, and causes changes in the behaviour of the target cells.

The nervous system contains large populations of non-neuronal cells, termed neuroglia or glia. These cells do not generate action potentials, but convey information encoded as transient changes in intracellular calcium concentration, termed calcium signalling. Glia interact with neurones in many different ways; their two-way communication is essential for normal brain activity.

It was thought for many years that glia outnumbered neurones by 10 times in the CNS, but recent studies using the isotropic fractionator method have challenged that popular view, suggesting instead that the two cell populations are rather similar in size (Azevedo et al 2009). That said, the glia : neurone ratio has been reported to be as high as 17 : 1 in the thalamus (Pakkenberg and Gundersen 1988).

The glial population in the CNS consists of microglia and macroglia; the latter are subdivided into oligodendrocytes and astrocytes. The principal glial cell in the PNS is the Schwann cell. Satellite cells surround each neuronal soma in ganglia.

For further reading on the nervous system, see Finger (2001), Kandel et al (2012), Kettenmann and Ransom (2012), Levitan and Kaczmarek (2001), Nicholls et al (2011) and Squire et al (2012).

NEURONES

Most of the neurones in the CNS are either clustered into nuclei, columns or layers, or dispersed within grey matter. Neurones in the PNS are confined to ganglia. Irrespective of location, neurones share many general features, which are discussed here in the context of central neurones. Special characteristics of ganglionic neurones and their adjacent tissues are discussed on page 57.

Neurones exhibit great variability in their size (cell bodies range from 5 to 100 μm diameter) and shape (Spruston 2008). Their surface areas are extensive because most neurones display numerous branched cell processes. They usually have a rounded or polygonal cell body (perikaryon or soma). This is a central mass of cytoplasm that encloses a nucleus and gives off long, branched extensions with which most intercellular contacts are made. Typically, one of these processes, the axon, is much longer than the others, the dendrites (**Fig. 3.2**). Generally, dendrites conduct electrical signals towards a soma whereas axons conduct impulses away from it.

Neurones can be classified according to the number and arrangement of their processes (Bota and Swanson 2007). Multipolar neurones (**Fig. 3.3**) are common; they have an extensive dendritic tree that arises either from a single primary dendrite or directly from the soma, and a single axon. Bipolar neurones, which typify neurones of the special sensory systems, have only a single dendrite that emerges from the soma opposite the axonal pole. Unipolar neurones, which transmit general sensation, e.g. dorsal root ganglion neurones, have a single short process that bifurcates into a peripheral and a central process. This arrangement arises by the fusion of the proximal axonal and dendritic processes of a bipolar neurone during development, and so such neurones may also be termed pseudounipolar. Neurones may also be classified according to whether their axons terminate locally on other neurones (interneurones), or transmit impulses over long distances, often to distinct territories via defined tracts (projection neurones).

Neurones are postmitotic cells and, with few exceptions, they are not replaced when lost.

SOMA

The plasma membrane of the soma is generally unmyelinated and is contacted by both inhibitory and excitatory axosomatic synapses; very occasionally, somasomatic and dendrosomatic contacts may be made. The non-synaptic surface may contain gap junctions and is partly covered by either astrocytic or satellite oligodendrocyte processes.

The cytoplasm of a typical soma (see Fig. 3.2) is rich in rough and smooth endoplasmic reticulum and free polyribosomes, indicating a high level of protein synthetic activity. Free polyribosomes often

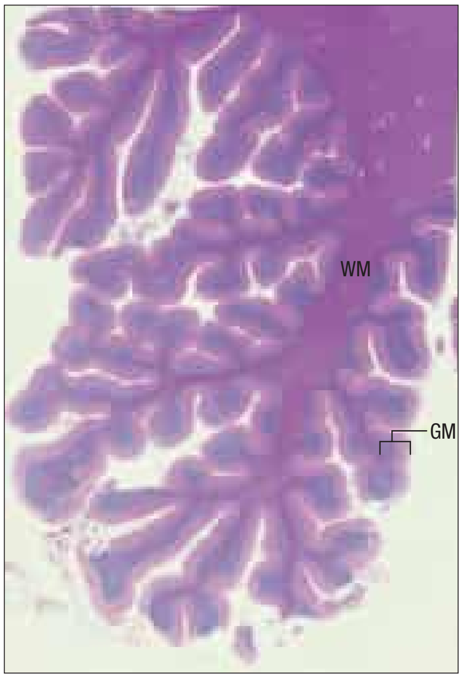

Fig. 3.1 A section through the human cerebellum stained to show the arrangement in the brain of the central white matter (WM, deep pink) and the highly folded outer grey matter (GM). In the cerebellum, GM consists of an inner granular layer of tightly packed small neurones (blue) and an outermost molecular layer (pale pink), where neuronal processes make synaptic contacts. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

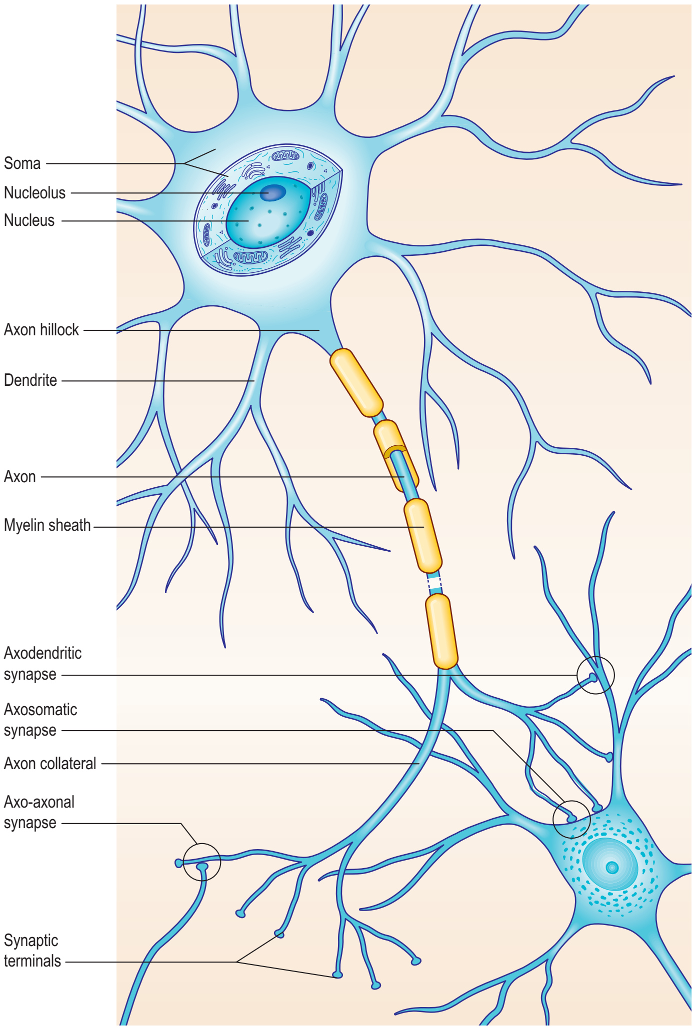

Fig. 3.2 A schematic view of typical neurones featuring one with the soma cut away to show the nucleus and cytoplasmic organelles, dendritic trees with synaptic contacts, other types of synapse, the axon hillock and a myelinated axon.

congregate in large groups associated with the rough endoplasmic reticulum. These aggregates of RNA-rich structures are visible by light microscopy as basophilic Nissl bodies or granules. They are distributed throughout the cell body and large dendrites; the axon hillock is conspicuously ribosome-free. Nissl bodies are more obvious in large, highly active cells, such as spinal motor neurones (**Fig. 3.4**), which contain large stacks of rough endoplasmic reticulum and polyribosome aggregates. Maintenance and turnover of cytoplasmic and membranous components are necessary activities in all cells; the huge total volume of cytoplasm within the soma and processes of many neurones requires a considerable commitment of protein synthetic machinery. Neurones also synthesize other proteins (enzyme systems, G-protein coupled receptors, scaffold proteins) involved in the production of neurotransmitters and in the reception and transduction of incoming stimuli. Various transmembrane channel proteins and enzymes are located at the surfaces of neurones, where they are associated with movements of ions.

The nucleus is characteristically large and euchromatic, and contains at least one prominent nucleolus; these are features typical of all cells engaged in substantial levels of protein synthesis. The cytoplasm contains many mitochondria and moderate numbers of lysosomes. Golgi complexes are usually close to the nucleus, near the bases of the main dendrites and opposite the axon hillock.

The neuronal cytoskeleton is a prominent feature of its cytoplasm and gives shape, strength and support to the dendrites and axons. A number of neurodegenerative diseases are characterized by abnormal aggregates of cytoskeletal proteins (reviewed in Cairns et al (2004)). Neurofilaments (the intermediate filaments of neurones) and microtubules are abundant in the soma and along dendrites and axons; the proportions vary with the type of neurone and cell process. Bundles of neurofilaments constitute neurofibrils, which can be seen by light microscopy in silver-stained or immunolabelled sections. Neurofilaments are heteropolymers of proteins assembled from three polypeptide subunits, NF-L (68 kDa), NF-M (160 kDa) and NF-H (200 kDa). NF-M and NF-H have long C-terminal domains that project as side arms from the assembled neurofilament and bind to neighbouring filaments. They can be heavily phosphorylated, particularly in the highly stable neurofilaments of mature axons, and are thought to give axons their tensile strength. Some axons are almost filled by neurofilaments.

Microtubules are important in axonal transport, although dendrites usually have more microtubules than axons. Centrioles persist in mature postmitotic neurones, where they are concerned with the generation of microtubules rather than cell division. Centrioles are associated with cilia on the surfaces of developing neuroblasts. Their significance, other than at some sensory endings (e.g. the olfactory mucosa), is not known.

Pigment granules (**Fig. 3.5**) appear in certain regions, e.g. neurones of the substantia nigra contain neuromelanin, which is probably a waste product of catecholamine synthesis. A similar pigment gives a bluish colour to the neurones in the locus coeruleus. Some neurones are unusually rich in metals, which may form components of enzyme systems, e.g. zinc in the hippocampus and iron in the red nucleus. Ageing neurones, especially in spinal ganglia, accumulate granules of lipofuscin (senility pigment) in residual bodies, which are lysosomes packed with partially degraded lipoprotein material.

DENDRITES

Dendrites are highly branched, usually short processes that project from the soma (see Fig. 3.2; Shah et al 2010). The branching patterns of many dendritic arrays are probably established by random adhesive interactions between dendritic growth cones and afferent axons that occur during development. There is an overproduction of dendrites in early development, and this is pruned in response to functional demand as the nervous system matures and information is processed through the dendritic tree. There is evidence that dendritic trees may be plastic structures throughout adult life, expanding and contracting as the traffic of synaptic activity varies through afferent axodendritic contacts (for a review, see Wong and Ghosh (2002)). Groups of neurones with similar functions have a similar stereotypic tree structure (**Fig. 3.6**), suggesting that the branching patterns of dendrites are important determinants of the integration of the afferent inputs that converge on the tree. For a review of current research on dendritic trees in the normal and pathological brain, see Kulkarni and Firestein (2012).

Dendrites differ from axons in many respects. They represent the afferent rather than the efferent system of the neurone, and receive both excitatory and inhibitory axodendritic contacts. They may also make dendrodendritic and dendrosomatic connections (see Fig. 3.9), some of which are reciprocal. Synapses occur either on small projections called dendritic spines or on the smooth dendritic surface. Dendrites contain ribosomes, smooth endoplasmic reticulum, microtubules, neurofilaments, actin filaments and Golgi complexes. Their neurofilament proteins are poorly phosphorylated and their microtubules express the microtubule-associated protein (MAP)-2 almost exclusively in comparison with axons.

The shapes of dendritic spines range from simple protrusions to structures with a slender stalk and expanded distal end. Most spines are not more than 2 µm long, and have one or more terminal expansions; they can also be short and stubby, branched or bulbous. Large mushroom spines are assumed to have differentiated in response to afferent activity ('memory spines'; Matsuzaki et al 2004). These large spines often contain a spine apparatus, an organelle consisting of small sacs of endoplasmic reticulum interleaved by electron-dense bars (Gray 1959, Segal et al 2010). Mouse mutants deficient in these organelles show memory deficits (Deller et al 2003). Free ribosomes and polyribosomes are concentrated at the base of the spine. Ribosomal accumulations near synaptic sites provide a mechanism for activity-dependent synaptic plasticity through the local regulation of protein synthesis.

AXONS

The axon originates either from the soma or from the proximal segment of a dendrite at a specialized region free of Nissl granules, the axon hillock (see Fig. 3.2). Action potentials are initiated here, at the junction with the proximal axon (axon initial segment). The axonal plasma membrane (axolemma) is undercoated at the hillock by a concentration of cytoskeletal molecules, including spectrin and actin fibrils, which are important in anchoring numerous voltage-sensitive channels to the membrane. For details, see Bender and Trussell (2012), and for neural electrophysiological techniques, see Sakmann and Neher (2009). The

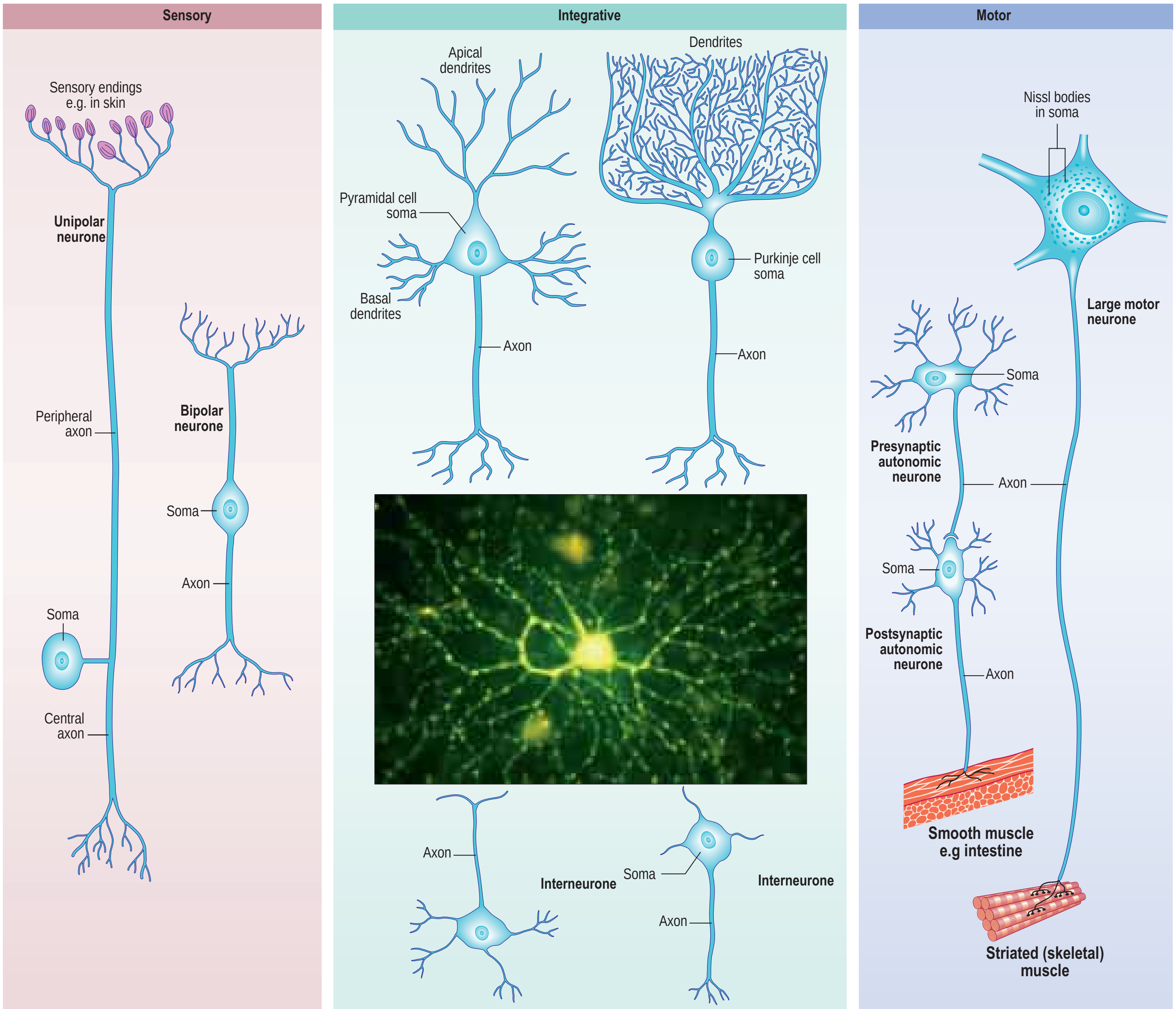

Fig. 3.3 The variety of shapes of neurones and their processes. The inset shows a human multipolar retinal ganglion cell, filled with fluorescent dye by microinjection. (Inset, Courtesy of Drs Richard Wingate, James Morgan and Ian Thompson, King's College, London.)

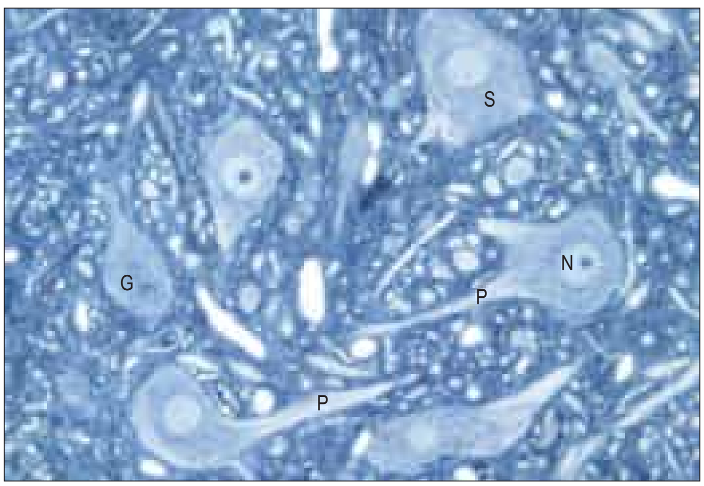

Fig. 3.4 Spinal motor neurones (toluidine blue stained resin section, rat tissue) showing a group of cell bodies (somata, S), some with the proximal parts of axonal and dendritic processes (P) visible. Their nuclei (N) typically have prominent, deeply staining nucleoli, indicative of metabolically highly active cells; two are visible in the plane of section. Nissl granules (G) are seen in the cytoplasm. Surrounding the neuronal somata is the neuropil, consisting of the interwoven processes of these and other neurones and of glial cells.

Fig. 3.5 Neurones in the substantia nigra of the human midbrain, showing cytoplasmic granules of neuromelanin pigment. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

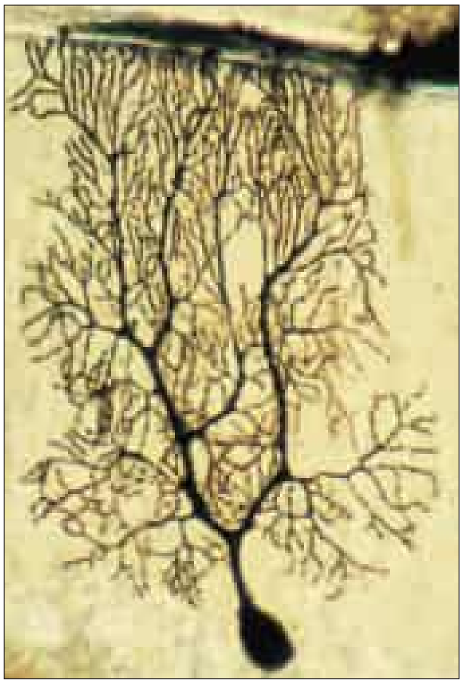

Fig. 3.6 A Purkinje neurone from the cerebellum of a rat stained by the Golgi–Cox method, showing the extensive two-dimensional array of dendrites. (Courtesy of Dr Martin Sadler and Professor M Berry, Division of Anatomy and Cell Biology, GKT School of Medicine, London.)

axon hillock is unmyelinated and often participates in inhibitory axo-axonal synapses. It is unique because it contains ribosomal aggregates immediately below the postsynaptic membrane (Kole and Stuart 2012).

In the CNS, small, unmyelinated axons lie free in the neuropil, whereas in the PNS they are embedded in Schwann cell cytoplasm. Myelin, which is formed around almost all axons of >2 µm diameter by oligodendrocytes in the CNS and by Schwann cells in the PNS, begins at the distal end of the axon hillock. Nodes of Ranvier are specialized regions of myelin-free axon where action potentials are generated and where an axon may branch. In both CNS and PNS, the territory of a myelinated axon between adjacent nodes is called an internode; the region close to a node, where the myelin sheath terminates, is called the paranode; and the region just beyond that is the juxtaparanode. Myelin thickness and internodal lengths are, in general, positively correlated with axon diameter. The density of sodium channels in the axolemma is highest at nodes of Ranvier, and very low along internodal membranes; sodium channels are spread more evenly within the axolemma of unmyelinated axons. Fast potassium channels are present in the paranodal regions of myelinated axons. Fine processes of glial cytoplasm (astrocytic in the CNS, Schwann cell in the PNS) surround the nodal axolemma.

The terminals of an axon are unmyelinated. Most expand into presynaptic boutons, which may form connections with axons, dendrites, neuronal somata or, in the periphery, muscle fibres, glands and lymphoid tissue. Exceptions include the free afferent sensory endings in, for example, the epidermis, which are unspecialized structurally, and the peripheral terminals of afferent sensory fibres with encapsulated endings (see Fig. 3.27). Axon terminals contain abundant small clear synaptic vesicles and large dense-core vesicles. The former contain a neurotransmitter (e.g. glutamate, γ-aminobutyric acid (GABA), acetylcholine) that is released into the synaptic cleft on the arrival of an action potential at the terminal and which then binds to cognate receptors on the postsynaptic membrane. Depending on the nature of the transmitter and its receptors, the postsynaptic neurone will become excited or inhibited. The dense-core vesicles contain neuropeptides, including brain-derived neurotrophic factor (BDNF; Dieni et al 2012). Axon terminals may themselves be contacted by other axons, forming axo-axonal presynaptic inhibitory circuits. Further details of neuronal microcircuitry are given in Kandel et al (2012) and Haines (2006).

Axons contain microtubules, neurofilaments, mitochondria, membrane vesicles, cisternae and lysosomes. They do not usually contain ribosomes or Golgi complexes, other than at the axon hillock; exceptionally, the neurosecretory fibres of hypothalamo-hypophysial neurones contain the mRNA of neuropeptides. Organelles are differentially distributed along axons, e.g. there is a greater density of mitochondria and membrane vesicles in the axon hillock, at nodes and in presynaptic endings. Axonal microtubules are interconnected by cross-linking MAPs, of which tau is the most abundant. Microtubules have an intrinsic polarity, and in axons all microtubules are uniformly orientated with their rapidly growing ends directed away from the soma towards the axon terminal. The microtubule binding protein tau plays an important role in Alzheimer's disease (Cairns et al 2004): formation of tau oligomers and the subsequent pathological filament arrays are critical steps in the aetiopathogenesis of this condition. Neurofilament proteins ranging from high to low molecular weights are highly phosphorylated in mature axons, whereas growing and regenerating axons express a calmodulin-binding membrane-associated phosphoprotein, growth-associated protein-43 (GAP-43), as well as poorly phosphorylated neurofilaments.

Neurones respond differently to injury, depending on whether the damage occurs in the CNS or the PNS. The glial microenvironment of a damaged central axon does not facilitate axonal regrowth; consequently, reconnection with original synaptic targets does not normally occur. In marked contrast, the glial microenvironment in the PNS is capable of facilitating axonal regrowth. However, functional outcome of clinical repair of a large mixed peripheral nerve, especially if the injury occurs some distance from the target organ, or produces a long defect in the damaged nerve, is frequently unsatisfactory (Birch 2011; see also Commentary 1.6).

Axoplasmic flow

Neuronal organelles and cytoplasm are in continual motion. Bidirectional streaming of vesicles along axons results in a net transport of materials from the soma to the terminals, with more limited movement in the opposite direction. Two major types of transport occur, one slow and one relatively fast. Slow axonal transport is a bulk flow of axoplasm only in the anterograde direction, carrying cytoskeletal proteins and soluble, non-membrane-bound proteins from the soma to the terminals at a rate of approximately 0.1–3 mm a day. In contrast, fast axonal transport carries membrane-bound vesicular material (endosomes and lysosomal autophagic vacuoles) and mitochondria at approximately 200 mm a day in the retrograde direction (towards the soma) and approximately 40 mm per day anterogradely (in particular, synaptic vesicles containing neurotransmitters).

Rapid flow depends on microtubules. Vesicles with side projections line up along microtubules and are transported along them by their side arms. Two microtubule-based motor proteins with adenosine 5′-triphosphatase (ATPase) activity are involved in fast transport: kinesin family proteins are responsible for the fast component of anterograde transport, and cytoplasmic dynein is responsible for retrograde transport. Retrograde transport mediates the movement of neurotrophic viruses, e.g. herpes zoster, rabies and polio, from peripheral terminals, and their subsequent concentration in the neuronal soma. It has been suggested that specific pools of endocytic organelles, signalling endosomes, mediate the long-distance axonal transport of growth factors, such as neurotrophins and their signalling receptors. Defects in axonal and dendritic transport have been linked to various neurodegenerative processes. See Guzik and Goldstein (2004), Hinckelmann et al (2013) and Schmieg et al (2014) for reviews of axonal transport in health and disease.

SYNAPSES

Transmission of impulses across specialized junctions (synapses) between two neurones is largely chemical and depends on the release of neurotransmitters from the presynaptic terminal. These neurotransmitters bind to cognate receptors in the postsynaptic neuronal membrane, resulting in a change of membrane conductance and leading to either a depolarization or a hyperpolarization (Ryan and Grant 2009).

The patterns of axonal termination vary considerably. A single axon may synapse with one neurone (e.g. climbing fibres ending on cerebellar Purkinje neurones), or more often with many neurones (e.g. cerebellar parallel fibres, which provide an extreme example of this phenomenon). In synaptic glomeruli (e.g. in the olfactory bulb), groups of synapses between two or many neurones form interactive units encapsulated by neuroglia (Fig. 3.7; Perea et al 2009).

Electrical synapses (direct communication via gap junctions) are rare in the human CNS and are confined largely to groups of neurones with tightly coupled activity, e.g. the inspiratory centre in the medulla. They will not be discussed further here.

Classification of chemical synapses

Chemical synapses have an asymmetric structural organization (Figs 3.8–3.9) in keeping with the unidirectional nature of their transmission. Typical chemical synapses share a number of important features. They all display an area containing a presynaptic density apposed to a

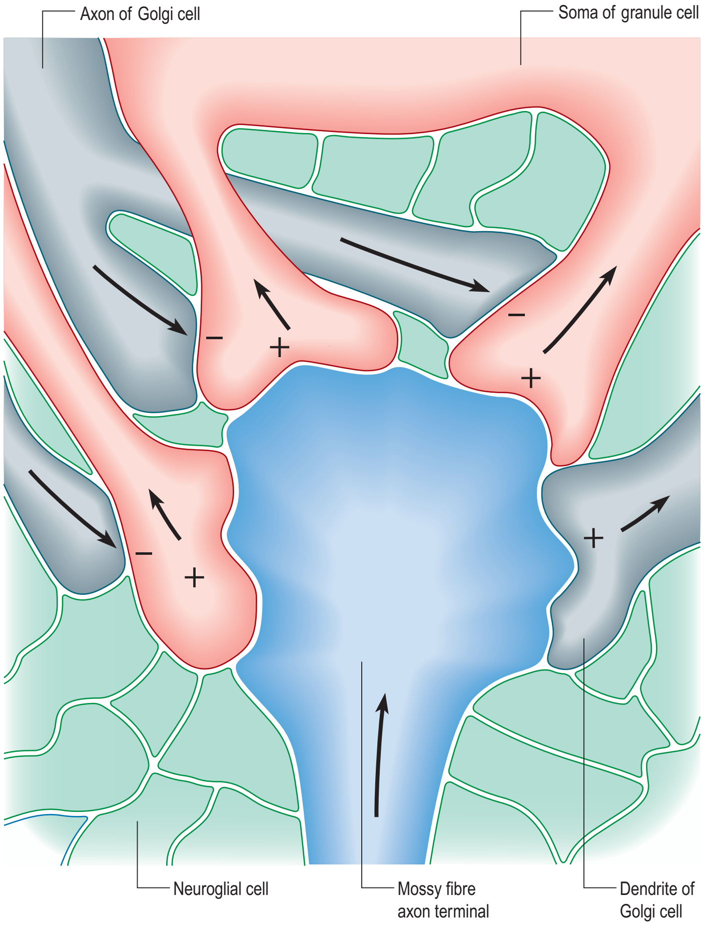

Fig. 3.7 The arrangement of a complex synaptic unit. A cerebellar synaptic glomerulus with excitatory ('+') and inhibitory ('–') synapses grouped around a central axonal bouton. The directions of transmission are shown by the arrows.

corresponding postsynaptic density; the two are separated by a narrow (20–30 nm) gap, the synaptic cleft. Synaptic vesicles containing the appropriate neurotransmitter are found on the presynaptic side, clustered near the presynaptic density at the presynaptic membrane. Together these define the active zone, the area of the synapse where neurotransmission takes place (Eggermann et al 2012, Gray 1959).

Chemical synapses can be classified according to a number of different parameters, including the neuronal regions forming the synapse, their ultrastructural characteristics, the chemical nature of their neurotransmitter(s) and their effects on the electrical state of the postsynaptic neurone. The following classification is limited to associations between neurones. Neuromuscular junctions share many (though not all) of these parameters, and are often referred to as peripheral synapses. They are described separately on page 63.

Synapses can occur between almost any surface regions of the participating neurones. The most common type occurs between an axon and either a dendrite or a soma, when the axon is expanded as a small bulb or bouton (see Figs 3.8–3.9). This may be a terminal of an axonal branch (terminal bouton) or one of a row of bead-like endings, when the axon makes contact at several points, often with more than one neurone (bouton de passage). Boutons may synapse with dendrites, including thorny protrusions named dendritic spines or the flat surface of a dendritic shaft; a soma (usually on its flat surface, but occasionally on spines); the axon hillock; and the terminal boutons of other axons.

The connection is classified according to the direction of transmission, and the incoming terminal region is named first. Most common are axodendritic synapses, although axosomatic connections are frequent. All other possible combinations are found but are less common, i.e. axo-axonal, dendro-axonal, dendrodendritic, somatodendritic or somatosomatic. Axodendritic and axosomatic synapses occur in all regions of the CNS and in autonomic ganglia, including those of the ENS. The other types appear restricted to regions of complex interaction between larger sensory neurones and microneurones, e.g. in the thalamus.

Ultrastructurally, synaptic vesicles may be internally clear or dense, and of different size (loosely categorized as small or large) and shape (round, flat or pleomorphic, i.e. irregularly shaped). The submembranous densities may be thicker on the postsynaptic than on the presynaptic side (asymmetric synapses), or equivalent in thickness (symmetrical synapses), and can be perforated or non-perforated. Synaptic ribbons

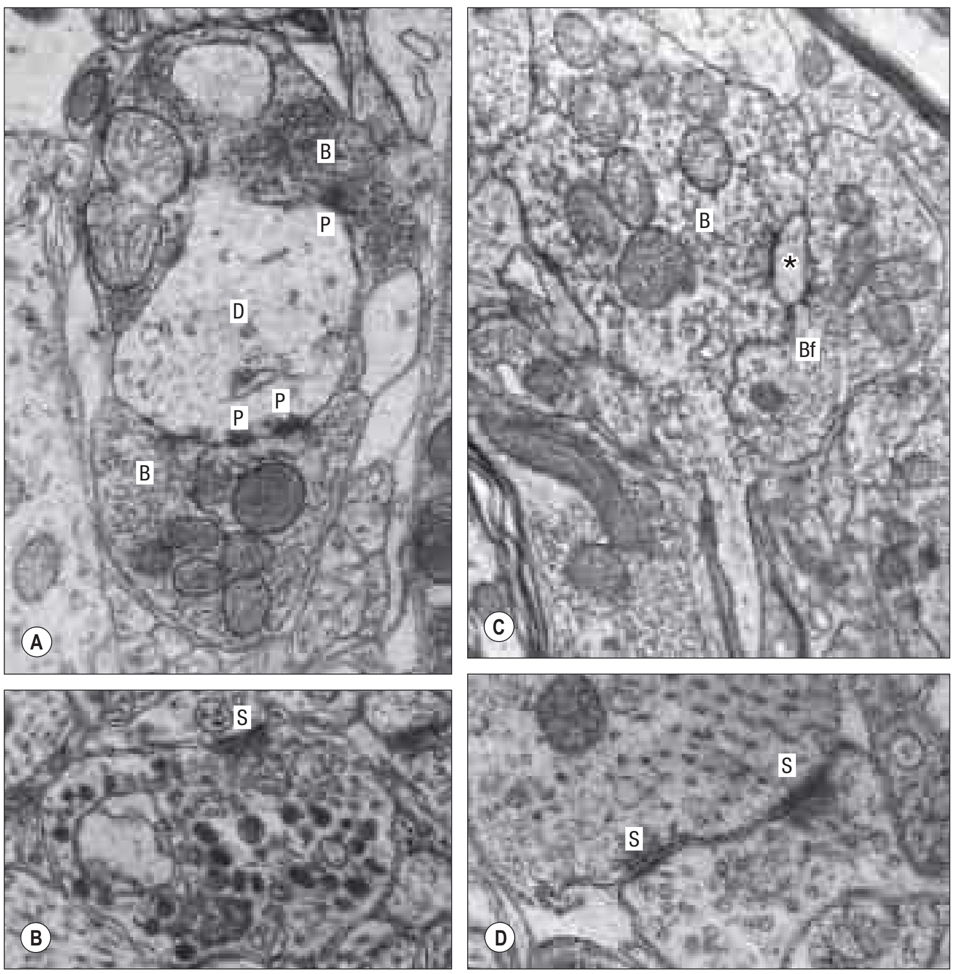

Fig. 3.8 Electron micrographs demonstrating various types of synapse. **A**, A cross-section of a dendrite (D) on which two synaptic boutons (B) end. The upper bouton contains round vesicles, and the lower bouton contains flattened vesicles of the small type. A number of pre- and postsynaptic (P) thickenings mark the specialized zones of contact. **B**, A type I synapse (S, postsynaptic site) containing both small, round, clear vesicles and also large, dense-cored vesicles of the neurosecretory type. **C**, A large terminal bouton (B) of an optic nerve afferent fibre, which is making contact with a number of postsynaptic processes, in the dorsal lateral geniculate nucleus of the rat. One of the postsynaptic processes (*) also receives a synaptic contact from a bouton (Bf) containing flattened vesicles. **D**, Reciprocal synapses (S) between two neuronal processes in the olfactory bulb. (Courtesy of Professor AR Lieberman, Department of Anatomy, University College, London.)

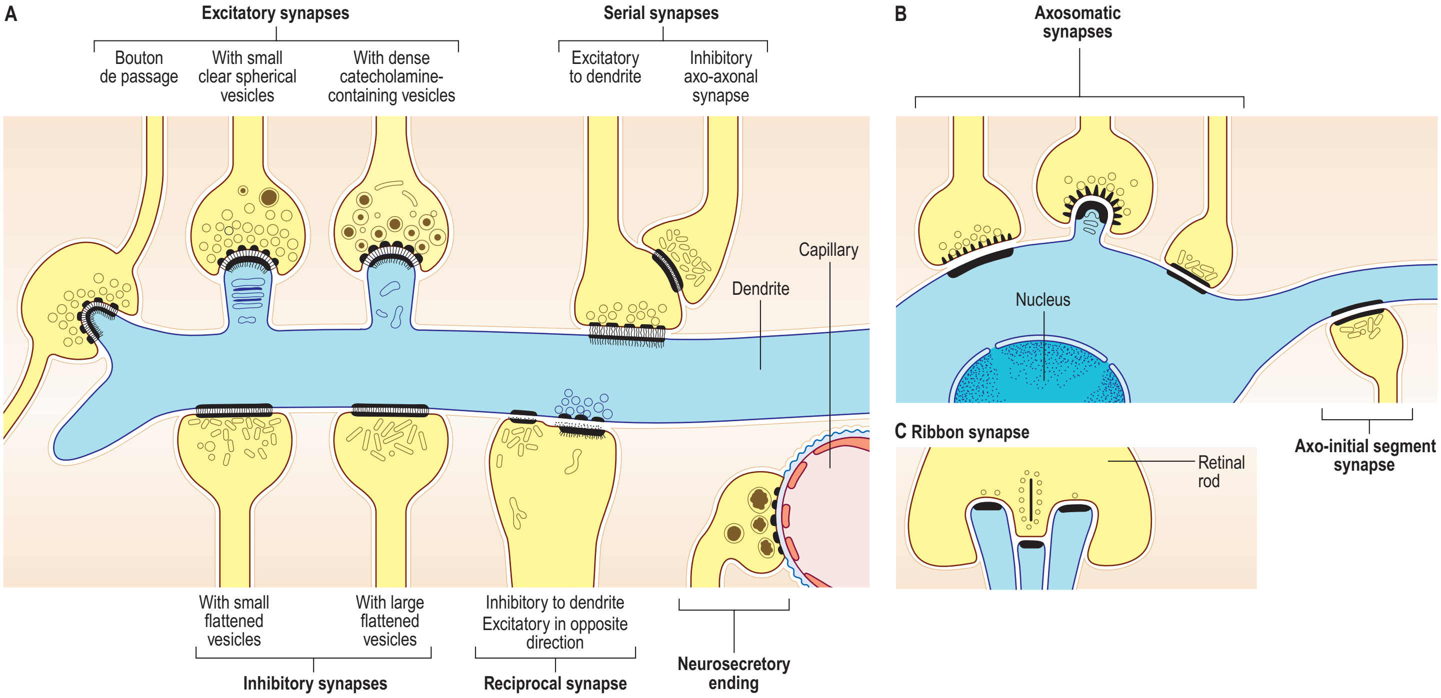

Fig. 3.9 The structural arrangements of different types of synaptic contact.

are found at sites of neurotransmission in the retina and inner ear. They have a distinctive morphology, in that the synaptic vesicles are grouped around a ribbon- or rod-like density orientated perpendicular to the cell membrane (see Fig. 3.9).

Synaptic boutons make obvious close contacts with postsynaptic structures but many other terminals lack specialized contact zones. Areas of transmitter release occur in the varicosities of unmyelinated axons, where effects are sometimes diffuse, e.g. the aminergic pathways of the basal ganglia, and in autonomic fibres in the periphery. In some instances, such axons may ramify widely throughout extensive areas of the brain and affect the behaviour of very large populations of neurones, e.g. the diffuse cholinergic innervation of the cerebral cortices. Pathological degeneration of these pathways can therefore cause widespread disturbances in neural function.

Neurones express a variety of neurotransmitters, either as one class of neurotransmitter per cell or more often as several. Good correlations exist between some types of transmitter and specialized structural features of synapses. In general, asymmetric synapses with relatively small spherical vesicles are associated with acetylcholine (ACh), glutamate, serotonin (5-hydroxytryptamine, 5-HT) and some amines; those with dense-core vesicles include many peptidergic synapses and other amines (e.g. noradrenaline (norepinephrine), adrenaline (epinephrine), dopamine). Symmetrical synapses with flattened or pleomorphic vesicles have been shown to contain either GABA or glycine.

Neurosecretory endings found in various parts of the brain and in neuroendocrine glands and cells of the dispersed neuroendocrine system share many features with presynaptic boutons. They all contain peptides or glycoproteins within dense-core vesicles. The latter are of characteristic size and appearance: they are often ellipsoidal or irregular in shape, and relatively large, e.g. oxytocin and vasopressin vesicles in the neurohypophysis may be up to 200 nm in diameter.

Synapses may cause depolarization or hyperpolarization of the postsynaptic membrane, depending on the neurotransmitter released and the classes of receptor molecule in the postsynaptic membrane. Depolarization of the postsynaptic membrane results in excitation of the postsynaptic neurone, whereas hyperpolarization has the effect of transiently inhibiting electrical activity. Subtle variations in these responses may also occur at synapses where mixtures of neuromediators are present and their effects are integrated. For details of the synaptic organization of the brain, see Shepherd (2003).

Type I and II synapses

There are two broad categories of synapse, type I and type II. In active zones of type I synapses the cytoplasmic density is thicker on the postsynaptic side. In type II synapses the pre- and postsynaptic densities are thinner and more symmetrical. Type I boutons contain a predominance of small spherical vesicles approximately 50 nm in diameter, and type II boutons contain oval or flattened vesicles. Throughout the CNS, type I synapses are generally excitatory and type II are inhibitory. In a few instances, types I and II synapses are found in close proximity, orientated in opposite directions across the synaptic cleft (a reciprocal synapse).

Mechanisms of synaptic activity

Synaptic activation begins with arrival of one or more action potentials at the presynaptic bouton, which causes the opening of voltage-sensitive calcium channels in the presynaptic membrane. The response time in typical fast-acting synapses is then very rapid; classic neurotransmitter (e.g. ACh, glutamate or GABA) is released in less than a millisecond. Release-ready synaptic vesicles are docked to the presynaptic membrane and primed through processes not yet fully understood. On Ca^{2+} influx through voltage-sensitive channels, their membranes fuse to open a pore through which neurotransmitter diffuses into the synaptic cleft (Eggermann et al 2012; Gray 1959).

Once the vesicle has discharged its contents, its membrane is incorporated into the presynaptic plasma membrane and is then recycled back into the bouton by endocytosis near the edges of the active zone. The recycling time for a synaptic vesicle may be in the range of a few seconds to minutes; newly recycled vesicles may be used instantly for the next cycle of neurotransmitter release (cycling pool of vesicles). The fusion of vesicles with the presynaptic membrane is responsible for the observed quantal behaviour of neurotransmitter release, both during neural activation and spontaneously, in the slightly leaky resting condition (Neher and Sakaba 2008; Suedhof 2012).

Postsynaptic events vary greatly, depending on the receptor molecules and their related molecular complexes (Murakoshi and Yasuda 2012). Receptors are generally classed as either ionotropic or metabotropic. Ionotropic receptors are multimeric protein complexes that harbour intrinsic ion channels that can be operated by conformational changes induced when neurotransmitter molecules bind the receptor complex, causing a voltage change within the postsynaptic cell. Examples are the nicotinic ACh receptor and the related $GABA_A$ receptor, which are formed from five subunits, and the tetrameric ionotropic glutamate receptors, such as the *N*-methyl-D-aspartate (NMDA) receptor or the α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor. Alternatively, the receptor and ion channel may be separate molecules, coupled by G-proteins, some via a complex cascade of chemical interactions (a second messenger system), e.g. the adenylate cyclase pathway. Postsynaptic effects are generally rapid and short-lived, because the transmitter is quickly inactivated either by an extracellular enzyme (e.g. acetylcholinesterase, AChE), or by uptake into neurones or glial cells. Examples of such metabotropic receptors are the muscarinic ACh receptor and the dopamine receptor.

Neurohormones

Neurohormones are included in the class of molecules with neurotransmitter-like activity. They are synthesized in neurones and released into the blood circulation by exocytosis at synaptic bouton-like structures. As with classic endocrine gland hormones, they may act at great distances from their site of secretion. Neurones secrete into the CSF or local interstitial fluid to affect other cells, either diffusely or at a distance. To encompass this wide range of phenomena the general term neuromediation has been used, and the chemicals involved are called neuromediators.

Neuromodulators

Some neuromediators do not appear to affect the postsynaptic membrane directly but they can affect its responses to other neuromediators, either enhancing their activity (by increasing or prolonging the immediate response), or perhaps limiting or inhibiting their action. These substances are called neuromodulators. A single synaptic terminal may contain one or more neuromodulators in addition to a neurotransmitter, usually (though not always) in separate vesicles. Neuropeptides (see below) are nearly all neuromodulators, at least in some of their actions. They are stored within dense granular synaptic vesicles of various sizes and appearances.

Development and plasticity of synapses

Embryonic synapses first appear as inconspicuous dense zones flanking synaptic clefts. Immature synapses often appear after birth, suggesting that they may be labile, and are reinforced if transmission is functionally effective, or withdrawn if redundant. This is implicit in some theories of memory (Squire and Kandel 2008), which postulate that synapses are modifiable by frequency of use, to establish preferential conduction pathways. Evidence from hippocampal neurones suggests that even brief synaptic activity can increase the strength and sensitivity of the synapse for some hours or longer (long-term potentiation, LTP). During early postnatal life, the normal developmental increase in numbers and sizes of synapses and dendritic spines depends on the degree of neural activity and is impaired in areas of damage or functional deprivation.

Neurotransmitter molecules

Until recently, the molecules known to be involved in chemical synapses were limited to a fairly small group of classic neurotransmitters, e.g. ACh, noradrenaline (norepinephrine), adrenaline (epinephrine), dopamine and histamine, all of which had well-defined rapid effects on other neurones, muscle cells or glands. However, many synaptic interactions cannot be explained on the basis of classic neurotransmitters, and it is now known that other substances, particularly some amino acids such as glutamate, glycine, aspartate, GABA and the monoamine, serotonin, also function as transmitters. Substances first identified as hypophysial hormones or as part of the dispersed neuroendocrine system (see below) of the alimentary tract, can be detected widely throughout the CNS and PNS, often associated with functionally integrated systems. Many of these are peptides; more than one hundred (together with other candidates) function mainly as neuromodulators and influence the activities of classic transmitters.

Acetylcholine

Acetylcholine (ACh) is perhaps the most extensively studied neurotransmitter of the classic type. Its precursor, choline, is synthesized in the neuronal soma and transported to the axon terminals, where it is acetylated by the enzyme choline acetyl transferase (ChAT), and stored in clear spherical vesicles 40–50 nm in diameter. ACh is synthesized by motor neurones and released at all their motor terminals on skeletal muscle. It is released by preganglionic fibres at synapses in parasympathetic and sympathetic ganglia, and many parasympathetic, and some sympathetic, ganglionic neurones are cholinergic. ACh is also associated with the degradative extracellular enzyme AChE, which inactivates the transmitter by converting it to choline.

The effects of ACh on nicotinic receptors (i.e. those in which nicotine is an agonist) are rapid and excitatory. In the CNS, the nicotinic ACh receptor mediates the effect of tobacco (for review, see Albuquerque et al (2009)). In the peripheral autonomic nervous system, the slower, more sustained excitatory effects of cholinergic autonomic endings are mediated by muscarinic receptors via a second messenger system.

Monoamines

Monoamines include the catecholamines (noradrenaline (norepinephrine), adrenaline (epinephrine) and dopamine), the indoleamine serotonin (5-hydroxytryptamine, 5-HT) and histamine (Haas et al 2008). They are synthesized by neurones in sympathetic ganglia and by their homologues, the chromaffin cells of the suprarenal medulla and paraganglia. Within the CNS, the somata of monoaminergic neurones lie mainly in the brainstem, although their axons ramify widely into all parts of the nervous system. Monoaminergic cells are also present in the retina.

Noradrenaline is the chief transmitter present in sympathetic ganglionic neurones with endings in various tissues, notably smooth muscle and glands, and in other sites including adipose and haemopoietic tissues and the corneal epithelium. It is also found at widely distributed synaptic endings within the CNS, many of them the terminals of neuronal somata situated in the locus coeruleus in the medullary floor. The actions of noradrenaline depend on its site of action and vary with the type of postsynaptic receptor, e.g. it strongly inhibits neurones of the submucosal plexus of the intestine and of the locus coeruleus via α_2-adrenergic receptors, whereas it mediates depolarization, producing vasoconstriction, via β-receptors in vascular smooth muscle. Adrenaline is present in central and peripheral nervous pathways and occurs with noradrenaline in the suprarenal medulla. Both adrenaline and noradrenaline are found in dense-cored synaptic vesicles approximately 50 nm in diameter.

Dopamine is a neuromediator of considerable clinical importance, found mainly in neurones with cell bodies in the telencephalon, diencephalon and mesencephalon. A major dopaminergic neuronal population in the midbrain constitutes the substantia nigra, so called because its cells contain neuromelanin, a black granular by-product of dopamine synthesis. Dopaminergic endings are particularly numerous in the corpus striatum, limbic system and cerebral cortex. Structurally, dopaminergic synapses contain numerous dense-cored vesicles that resemble those containing noradrenaline. Pathological reduction in dopaminergic activity has widespread effects on motor control, affective behaviour and other neural activities, as seen in Parkinson's syndrome.

Serotonin and histamine are found in neurones mainly within the CNS. Serotonin is typically synthesized in small midline neuronal clusters in the brainstem, mainly in the raphe nuclei; the axons from these neurones ramify extensively throughout the entire brain and spinal cord. Synaptic terminals contain rounded, clear vesicles approximately 50 nm in diameter and are of the asymmetrical type. Histaminergic neurones appear to be relatively sparse and are restricted largely to the hypothalamus.

Amino acids

There are three major amino acids: GABA, glutamate and glycine, which bind to specific receptors (Barrera and Edwardson 2008). GABA is a major inhibitory transmitter released at the terminals of local circuit neurones within the brainstem and spinal cord (e.g. the recurrent inhibitory Renshaw loop), cerebellum (where it is the main transmitter of Purkinje cells), basal ganglia, cerebral cortex, thalamus and subthalamus. It is stored in flattened or pleomorphic vesicles within symmetrical synapses. GABA may be inhibitory to postsynaptic neurones, or may mediate either presynaptic inhibition or facilitation, depending on the synaptic arrangement (Gassmann and Bettler 2012).

Glutamate is the major excitatory transmitter present widely within the CNS, including the major projection pathways from the cortex to the thalamus, tectum, substantia nigra and pontine nuclei. It is found in the central terminals of the auditory and trigeminal nerves, and in the terminals of parallel fibres ending on Purkinje cells in the cerebellum. Structurally, glutamate is associated with asymmetrical synapses containing small (approximately 30 nm), round, clear synaptic vesicles (Contractor et al 2011). For further reading, see Willard and Koochekpour (2013).

Glycine is a well-established inhibitory transmitter of the CNS, particularly the lower brainstem and spinal cord, where it is mainly found in local circuit neurones. Recent evidence suggests that glycine may also be released from glutamatergic axon terminals to participate in activation of NMDA receptors, and from astrocytes into the synaptic cleft via activation of non-NMDA-type glutamatergic ionotropic receptors in the glial cell membrane (see Harsing and Matyus (2013) for further references).

ATP and adenosine

ATP serves not only as a universal energy substrate, but also as an extracellular signalling molecule. Specific ionotropic (P2X) and metabotropic (P2Y) receptors are expressed in neurones and even more prominently on all types of glial cell. Adenosine is a degradation product of ATP and has specific metabotropic receptors that may be located presynaptically (Burnstock et al 2011).

Nitric oxide

Nitric oxide (NO) is of considerable importance at autonomic and enteric synapses, where it mediates smooth muscle relaxation. It functions in several types of synaptic plasticity, including hippocampal long-term potentiation (LTP), when it may act as a retrograde messenger after postsynaptic NMDA receptor activation. NO is able to diffuse freely through cell membranes, and so is not under such tight quantal control as vesicle-mediated neurotransmission.

Neuropeptides

Many neuropeptides coexist with other neuromediators in the same synaptic terminals. As many as three peptides often share a particular ending with a well-established neurotransmitter, in some cases within the same synaptic vesicles. Some peptides occur in both the CNS and the PNS, particularly in the ganglion cells and peripheral terminals of the ANS, whilst others are entirely restricted to the CNS. Only a few examples are given here.

Most of the neuropeptides are classified according to the site where they were first discovered. For example, the gastrointestinal peptides were initially found in the gut wall, and a group that includes releasing hormones, adenohypophysial and neurohypophysial hormones was first associated with the pituitary gland. Some of these peptides are closely related to each other in their chemistry because they are derived from the same gene products (e.g. the pro-opiomelanocortin group), which are cleaved to produce smaller peptides.

Substance P (SP) was the first of the peptides to be characterized as a gastrointestinal neuromediator and is considered to be the prototypic neuropeptide. It is an 11-amino-acid polypeptide that belongs to the tachykinin neuropeptide family, and is a major neuromediator in the brain and spinal cord. Contained within large granular synaptic vesicles, SP is found in approximately 20% of dorsal root and trigeminal ganglion cells, in particular in small nociceptive neurones, and in some fibres of the facial, glossopharyngeal and vagal nerves. Within the CNS, SP is present in several apparently unrelated major pathways, and has been described in the limbic system, basal ganglia, amygdala and hypothalamus. Its known action is prolonged postsynaptic excitation, particularly from nociceptive afferent terminals, which sustains the effects of noxious stimuli. SP is one of the main neuropeptides that trigger an inflammatory response in the skin and has also been implicated in the vomiting reflex, changes in cardiovascular tone, stimulation of salivary secretion, smooth muscle contraction, and vasodilation.

Vasoactive intestinal polypeptide (VIP), another gastrointestinal peptide, is widely present in the CNS, where it is probably an excitatory neurotransmitter or neuromodulator. It is found in distinctive bipolar neurones of the cerebral cortex; small dorsal root ganglion cells, particularly of the sacral region; the median eminence of the hypothalamus, where it may be involved in endocrine regulation; intramural ganglion cells of the gut wall; and sympathetic ganglia.

Somatostatin (ST, somatotropin release inhibiting factor) has a broad distribution within the CNS, and may be a central neurotransmitter or neuromodulator. It also occurs in small dorsal root ganglion cells. Beta-endorphin, leu- and metenkephalins, and the dynorphins belong to a group of peptides called the naturally occurring opiates that possess analgesic properties. They bind to opiate receptors in the brain where, in general, their action seems to be inhibitory. Enkephalins have been localized in many areas of the brain. Their particular localization in the septal nuclei, amygdaloid complex, basal ganglia and hypothalamus suggests that they are important mediators in the limbic system and in the control of endocrine function. They have also been implicated strongly in the central control of pain pathways, because they are found in the peri-aqueductal grey matter of the midbrain, a number of reticular raphe nuclei, the spinal nucleus of the trigeminal nerve and the substantia gelatinosa of the spinal cord. The enkephalinergic pathways exert an important presynaptic inhibitory action on nociceptive afferents in the spinal cord and brainstem. Like many other neuromediators, enkephalins also occur widely in other parts of the brain in lower concentrations.

CENTRAL GLIA

Glial (neuroglial) cells (Fig 3.10) vary considerably in type and number in different regions of the CNS. There are two major groups, macroglia (astrocytes and oligodendrocytes) and microglia, classified according to origin. Macroglia arise within the neural plate, in parallel with neurones, and constitute the great majority of glial cells. Their functions are diverse and are now known to extend beyond a passive supporting role (reviewed in Kettenmann and Ransom (2012)). Microglia have a small soma (see Fig. 3.19) and are derived from a distinct lineage of monocytic cells originating from the yolk sac.

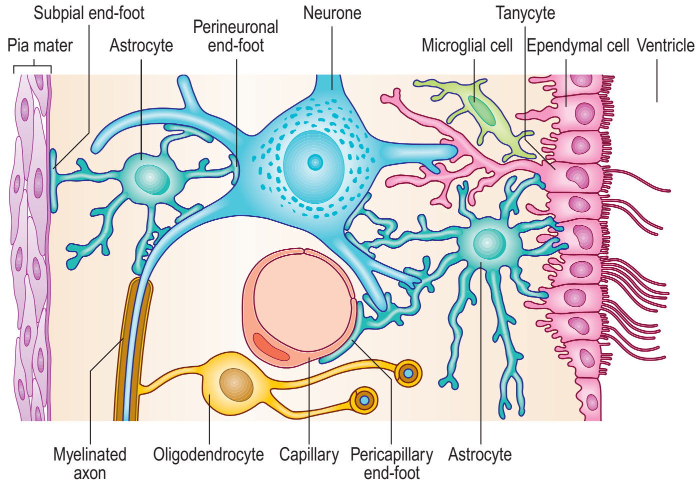

Fig. 3.10 The different types of non-neuronal cell in the CNS and their structural organization and interrelationships with each other and with neurones.

ASTROCYTES

Astrocytes are the most abundant and diverse glial cell type but their identity is not well defined (Matyash and Kettenmann 2010). There is no common marker that labels all astrocytes, in the way that myelin basic protein is a marker for oligodendrocytes or the calcium-binding protein Iba1 is a marker for microglia. A commonly used marker is the expression of glial fibrillary acidic protein (GFAP), which forms intermediate filaments, but GFAP is not expressed in all astrocytes.

The morphology of astrocytes is extremely diverse. Classically, two forms were distinguished: protoplasmic and fibrous astrocytes. Protoplasmic astrocytes (star-shaped cells) are found in grey matter, possess several stem processes that branch further into a very complex network, and contact synapses, both at the pre- and postsynaptic membranes. Fibrous astrocytes are predominantly found in white matter and their processes are often orientated in parallel with the axons. Radial glial cells are found early in development and serve as stem cells for neurones and glial cells. They may be categorized as astrocytes because they transform later in development into typical astrocytes. There are a number of other types of astrocyte with specialized morphologies. Bergmann glial cells in the cerebellum have somata in the Purkinje cell layer, processes that intermingle with the dendritic trees of the Purkinje neurones and terminal end-feet at the pial surface. Müller cells in the retina have a radial morphology and span the entire retina. Other astrocytic cells are tanycytes, velate astrocytes (cerebellum) and pituicytes (infundibulum and neurohypophysis of the pituitary gland). Pituicyte processes end mostly on endothelial cells in the neurohypophysis and tuber cinereum.

Astrocyte complexity and morphological diversity has reached the highest evolutionary level in humans (Fig. 3.11). A single astrocyte may enwrap several neuronal somata and make contacts with tens of thousands of individual synapses; bipolar astrocytes located in layer 5 and 6 of the cortex may extend processes up to 1 mm long.

Astrocytes in grey matter form a syncytium in which cells are interconnected by gap junctions, permitting the exchange of ions (e.g. calcium, propagated in waves) and small molecules such as ATP or glucose. They are the only cells in the brain capable of converting glucose into glycogen, which serves as an energy store. Before re-releasing glucose, astrocytes convert it to lactate, which is taken up by neurones; failure in glucose flow through the astrocytic network results in impairment of neuronal function.

Astrocytes not only respond to neuronal activity but also modulate that activity. They enwrap all penetrating and intracerebral arterioles and capillaries, control the ionic and metabolic environment of the neuropil and mediate neurovascular coupling. They form specialized structures that contact either the pial surface (as the glia limitans) or blood vessels; their end-feet entirely enwrap blood vessels and are instrumental in the induction of the blood–brain barrier.

Traumatic injury to the CNS induces astrogliosis, seen as a local increase in the number and size of GFAP-positive cells and a characteristic extensive meshwork of processes. The microenvironment of this glial scar, which may also include cells of oligodendrocyte lineage and myelin debris, plays an important role in inhibiting regrowth of damaged CNS axons (Robel et al 2011, Seifert et al 2006).

Astrocytes control the diameter of the vessels they contact and can trigger either their dilation or their contraction, depending on the substances they release and the levels of associated neuronal activity. They express water channels (aquaporins) at the end-feet covering the capillaries; it has been suggested that this may represent the means by which astrocytes control brain volume (Tait et al 2008), and it may be relevant to understanding mechanisms of brain tissue swelling, a major clinical complication. Astrocytes express different glutamate transporters that efficiently maintain low levels of extracellular glutamate, which is excitotoxic. Internalized glutamate is converted into glutamine and released from astrocytes to be taken up by local neurones and reconverted to glutamate via the glutamate–glutamine cycle. They play a similar role in controlling extracellular GABA levels via expression of GABA transporters. Astrocytes possess both passive and active mechanisms to control extracellular potassium levels at a resting level of about 3 mmol. They also express transporters that regulate pH and are thought to play an important role in controlling extracellular pH in the brain. For further reading on the concept of the 'tripartite synapse', where astrocytic processes interact with pre- and postsynaptic neuronal elements, see Haydon and Carmignoto (2006).

It has become evident that astrocytes are involved in the modulation of long-term potentiation (considered as a cellular mechanism of memory formation) and heterosynaptic depression. They modulate neuronal activity by releasing neuroactive substances such as D-serine, ATP or glutamate; it is unclear whether they express all the elements required for neurotransmitter release by a vesicular mechanism (Parpura and Zorec 2010).

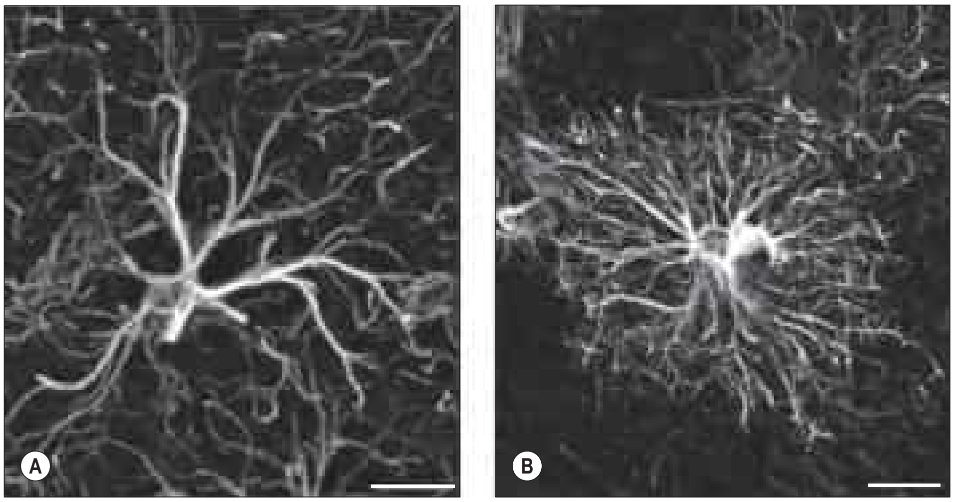

Fig. 3.11 Human protoplasmic astrocytes are larger and more complicated than their rodent counterparts. **A**, A typical mouse protoplasmic astrocyte. Glial fibrillary acidic protein (GFAP) immunostain; white. SB=20 μm. **B**, A typical human protoplasmic astrocyte to the same scale. SB=20 μm. (From Oberheim NA, Takano T, Han X, et al 2009 Uniquely hominid features of adult human astrocytes. J Neurosci 29:3276–87.)

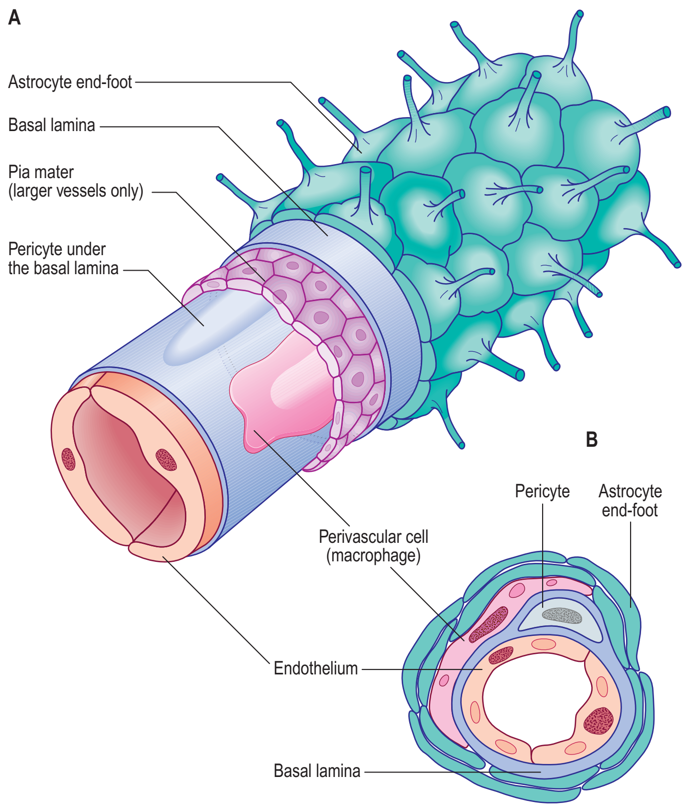

Fig. 3.12 The relationship between the glia limitans, perivascular cells and blood vessels within the brain, in longitudinal (**A**) and transverse (**B**) sections. A sheath of astrocytic end-feet wraps around the vessel and, in vessels larger than capillaries, its investment of pial meninges. Vascular endothelial cells are joined by tight junctions and supported by pericytes; perivascular macrophages lie outside the endothelial basal lamina (light blue).

Blood–brain barrier

Proteins circulating in the blood enter most tissues of the body except those of the brain, spinal cord and peripheral nerves. This concept of a blood–brain or a blood–nerve barrier applies to many substances – some are actively transported across the blood–brain barrier, others are actively excluded. The blood–brain barrier is located at the capillary endothelium within the brain and is dependent on the presence of tight junctions (occluding junctions, zonulae adherentes) between endothelial cells coupled with a relative lack of transcytotic vesicular transport. The tightness of the barrier is substantially supported by the close apposition of astrocytes, which direct the formation of endothelial tight junctions, to blood capillaries (reviewed in Abbott et al (2006), Cardoso et al (2010); **Fig. 3.12**).

The blood–brain barrier develops during embryonic life but may not be fully completed by birth. There are certain areas of the adult brain where the endothelial cells are not linked by tight junctions, which means that a free exchange of molecules occurs between blood and adjacent brain. Most of these areas are situated close to the ventricles and are known as circumventricular organs; these areas make up less than 1% of the total area of the brain. Elsewhere, unrestricted diffusion through the blood–brain barrier is only possible for substances that can cross biological membranes because of their lipophilic character. Lipophilic molecules may be actively re-exported by the brain endothelium.

Breakdown of the blood–brain barrier occurs when the brain is damaged by ischaemia or infection, and is also associated with primary and metastatic cerebral tumours. Reduced blood flow to a region of the brain alters the permeability and regulatory transport functions of the barrier locally; the increased stress on compromised endothelial cells results in leakage of fluid, ions, serum proteins and intracellular substances into the extracellular space of the brain. The integrity of the barrier can be evaluated clinically using computed tomography and functional magnetic resonance imaging. Breakdown of the blood–brain barrier may be seen at postmortem in jaundiced patients who have had an infarction. Normally, the brain, spinal cord and peripheral nerves remain unstained by the bile post mortem, although the choroid plexus is often stained a deep yellow. However, areas of recent infarction (1–3 days) will also be stained by bile pigment because of the localized breakdown of the blood–brain barrier.

OLIGODENDROCYTES

Oligodendrocytes myelinate CNS axons and are most commonly seen as intrafascicular cells in myelinated tracts (**Figs 3.13–3.14**). They usually have round nuclei and their cytoplasm contains numerous mitochondria, microtubules and glycogen. They display a spectrum of morphological variation, from cells with large euchromatic nuclei and pale cytoplasm, to cells with heterochromatic nuclei and dense cytoplasm. In contrast to Schwann cells, which myelinate only one axonal segment, individual oligodendrocytes myelinate up to 50 axonal segments. Some oligodendrocytes are not associated with axons, and are either precursor cells or perineuronal (satellite) oligodendrocytes with processes that ramify around neuronal somata.

Within tracts, interfascicular oligodendrocytes are arranged in long rows interspersed at regular intervals with single astrocytes. Since oligodendrocyte processes are radially aligned to the axis of each row, myelinated tracts typically consist of cables of axons myelinated by a row of oligodendrocytes running down the axis of each cable.

Oligodendrocytes originate from the ventricular neurectoderm and the subependymal layer in the fetus, and continue to be generated from the subependymal plate postnatally. Stem cells migrate and seed into white and grey matter to form a pool of adult progenitor cells, which can later differentiate to replenish defunct oligodendrocytes, and possibly remyelinate axons in pathologically demyelinated regions. These cells display a highly branching morphology and express a specific chondroitin sulphate proteoglycan (Neuron Glia 2 (NG2) in rats and its homologue, melanoma cell surface chondroitin sulphate proteoglycan (MSCP), in humans). The name NG2 cell is used to describe the cells in both species: several different names have also been used since it was first recognized, including polydendrocyte (Nishiyama et al 2009) and syantocyte (Butt et al 2005). NG2 cells express a complex set of voltage-gated channels and ionotropic receptors for glutamate

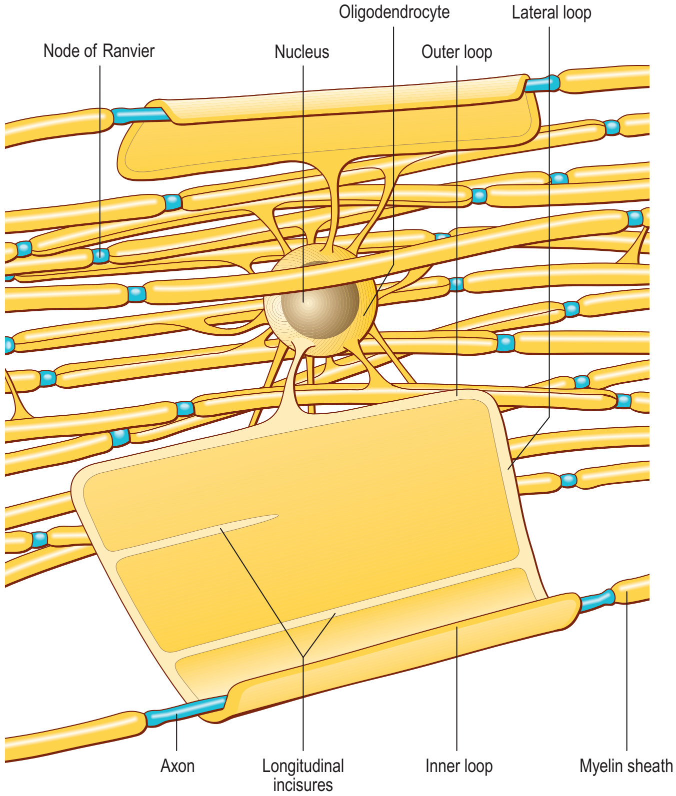

Fig. 3.13 The ensheathment of a number of axons by the processes of an oligodendrocyte. The oligodendrocyte soma is shown in the centre and its myelin sheaths are unfolded to varying degrees to show their extensive surface area. (Modified from Morell P, Norton WT (1980, May). Myelin, Scientific American, 242(5), 88–90, 92, 96 and Raine CS (1984), Morphology of Myelin and Myelination. In Myelin, 2nd ed. P Morell (ed) New York (Plenum Press), by permission.)

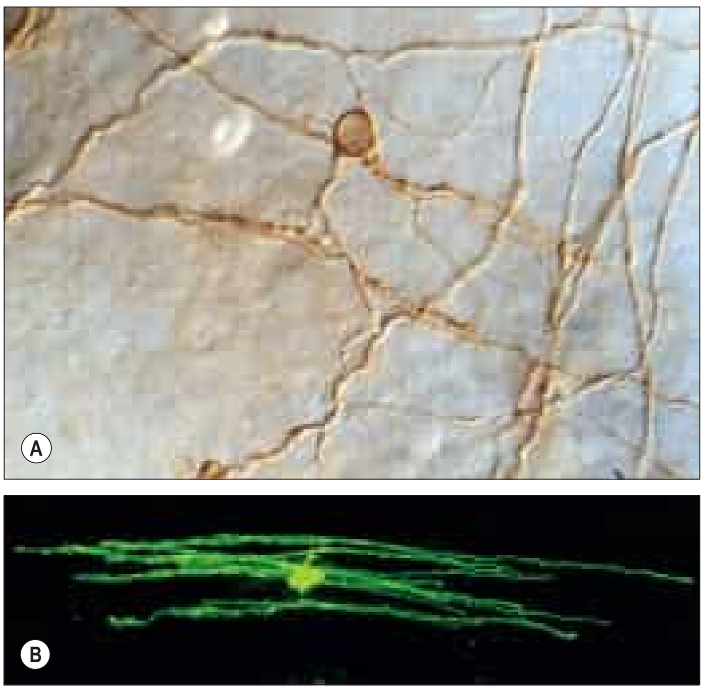

Fig. 3.14 **A**, An oligodendrocyte enwrapping several axons with myelin, demonstrated in a whole-mounted rat anterior medullary velum, immunolabelled with antibody to an oligodendrocyte membrane antigen. **B**, A confocal micrograph of a mature myelin-forming oligodendrocyte in an adult rat optic nerve, iontophoretically filled with an immunofluorescent dye by intracellular microinjection. (A, Courtesy of Fiona Ruge. B, Prepared by Professor A Butt, Portsmouth, and Kate Colquhoun, formerly Division of Physiology, GKT School of Medicine, London.)

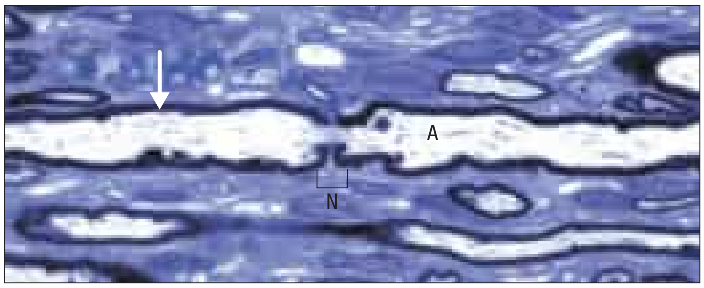

Fig. 3.15 A node of Ranvier (N) in the central nervous system of a rat. The pale-staining axon (A) is ensheathed by oligodendrocyte myelin (arrow), apart from a short, exposed region at the node. Toluidine blue stained resin section. (Courtesy of Dr Clare Farmer, King's College, London.)

and GABA; they form direct synapses with axons, enabling transient activation of these receptors (Hill and Nishiyama 2014). There is considerable support for the view that the NG2 cell is a distinct glial type.

Nodes of Ranvier and incisures of Schmidt–Lanterman

The territory ensheathed by an oligodendrocyte (or Schwann cell) process defines an internode, the interval between internodes is called a node of Ranvier (Fig. 3.15) and the territory immediately adjacent to the nodal gap is a paranode, where loops of oligodendrocyte cytoplasm abut the axolemma. Nodal axolemma is contacted by fine filopodia of perinodal cells, which have been shown in animal studies to have a presumptive adult oligodendrocyte progenitor phenotype; their function is unknown. Schmidt–Lanterman incisures are helical decompactions of internodal myelin where the major dense line of the myelin sheath splits to enclose a spiral of oligodendrocyte cytoplasm. Their structure suggests that they may play a role in the transport of molecules across the myelin sheath, but their function is not known.

MYELIN AND MYELINATION

Myelin is formed by oligodendrocytes (CNS) and Schwann cells (PNS). A single oligodendrocyte may ensheathe up to 50 separate axon segments, depending on calibre, whereas myelinating Schwann cells ensheathe axons on a 1 : 1 basis.

In general, myelin is laid down around axons above 2 µm in diameter. However, the critical minimal axon diameter for myelination is smaller and more variable in the CNS than in the PNS (approximately 0.2 µm in the CNS compared with 1–2 µm in the PNS). There is considerable overlap between the size of the smallest myelinated and the largest unmyelinated axons, and so axonal calibre is unlikely to be the only factor in determining myelination. Moreover, the first axons to become ensheathed ultimately attain larger diameters than those that are ensheathed at a later date. There is a reasonable linear relationship between axon diameter and internodal length and myelin sheath thickness: as the sheath thickens from a few lamellae to up to 200, the axon may also grow from 1 to 15 µm in diameter. Internodal lengths increase about 10-fold during the same time (Nave 2010).

It is not known precisely how myelin is formed in either PNS or CNS. Akt/mTOR (mammalian (or mechanistic) target of rapamycin) signalling has emerged as one of the major pathways involved in myelination; it has been implicated in the regulation of several steps during the development of myelinating Schwann cells and oligodendrocytes (Norrmén and Suter 2013). In the CNS, myelination also depends in part on expression of a protein (Wiskott–Aldrich syndrome protein family verprolin homologous; WAVE), which influences the actin cytoskeleton, oligodendrocyte lamellipodia formation and myelination (Kim et al 2006). The ultrastructural appearance of myelin is usually explained in terms of the spiral wrapping of an extensive, flat glial process (lamellipodium) around an axon, and the subsequent extrusion of cytoplasm from the sheath at all points other than incisures and paranodes. In this way, the compacted external surfaces of the plasma membrane of the ensheathing glial cell are thought to produce the minor dense lines, and the compacted inner cytoplasmic surfaces, the major dense lines, of the mature myelin sheath (Fig. 3.16). These lines, first described in early electron microscope studies of the myelin sheath, correspond to the intraperiod and period lines respectively, defined in X-ray studies of myelin. The inner and outer zones of occlusion of the spiral process are continuous with the minor dense line and are called

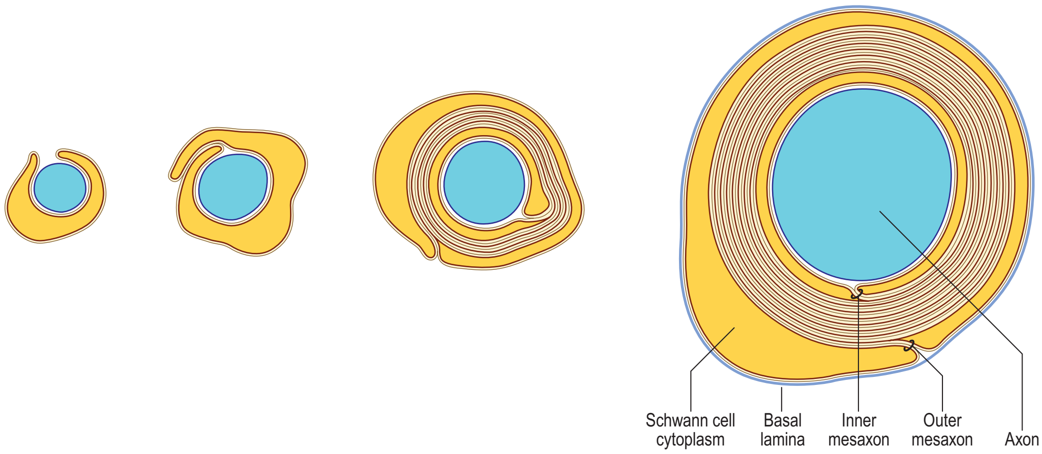

Fig. 3.16 Suggested stages in myelination of a peripheral axon by an ensheathing Schwann cell.

the inner and outer mesaxons. For further reading on aspects of myelination, see Bakhti et al (2013).

There are significant differences between central and peripheral myelin, reflecting the fact that oligodendrocytes and Schwann cells express different proteins during myelinogenesis. The basic dimensions of the myelin membrane are different. CNS myelin has a period repeat thickness of 15.7 nm whereas PNS myelin has a period to period line thickness of 18.5 nm, and the major dense line space is approximately 1.7 nm in CNS myelin, compared with 2.5 nm in PNS myelin. Myelin membrane contains protein, lipid and water, which forms at least 20% of the wet weight. It is a relatively lipid-rich membrane and contains 70–80% lipid. All classes of lipid have been found; perhaps not surprisingly, the precise lipid composition of PNS and CNS myelin is different. The major lipid species are cholesterol (the most common single molecule), phospholipids and glycosphingolipids. Minor lipid species include galactosylglycerides, phosphoinositides and gangliosides. The major glycolipids are galactocerebroside and its sulphate ester, sulphatide; these lipids are not unique to myelin but they are present in characteristically high concentrations. CNS and PNS myelin also contain low concentrations of acidic glycolipids, which constitute important antigens in some inflammatory demyelinating states. Gangliosides, which are glycosphingolipids characterized by the presence of sialic acid (*N*-acetylneuraminic acid), account for less than 1% of the lipid in myelin.

A relatively small number of protein species account for the majority of myelin protein. Some of these proteins are common to both PNS and CNS myelin, but others are different. Proteolipid protein (PLP) and its splice variant DM20 are found only in CNS myelin, whereas myelin basic protein (MBP) and myelin associated glycoprotein (MAG) occur in both. MAG is a member of the immunoglobulin supergene family, and is localized specifically at those regions of the myelin segment where compaction starts: namely, the mesaxons and inner periaxonal membranes, paranodal loops and incisures, in both CNS and PNS sheaths. It is thought to have a functional role in membrane adhesion.

In the developing CNS, axonal outgrowth precedes the migration of oligodendrocyte precursors, and oligodendrocytes associate with and myelinate axons after their phase of elongation; oligodendrocyte myelin gene expression is not dependent on axon association. In marked contrast, Schwann cells in the developing PNS are associated with axons during the entire phase of axonal growth. Myelination does not occur simultaneously in all parts of the body in late fetal and early postnatal development. White matter tracts and nerves in the periphery have their own specific temporal patterns that relate to their degree of functional maturity.

Mutations of the major myelin structural proteins have now been recognized in a number of inherited human neurological diseases. As would be expected, these mutations produce defects in myelination and in the stability of nodal and paranodal architecture that are consistent with the suggested functional roles of the relevant proteins in maintaining the integrity of the myelin sheath.

EPENDYMA

Ependymal cells line the ventricles (**Fig. 3.17**; see Fig. 3.10) and central canal of the spinal cord. They form a single-layered epithelium that varies from squamous to columnar in form. At the ventricular surface, cells are joined by gap junctions and occasional desmosomes. Their apical surfaces have numerous microvilli and/or cilia, the latter contributing to the flow of CSF. There is considerable regional variation in the

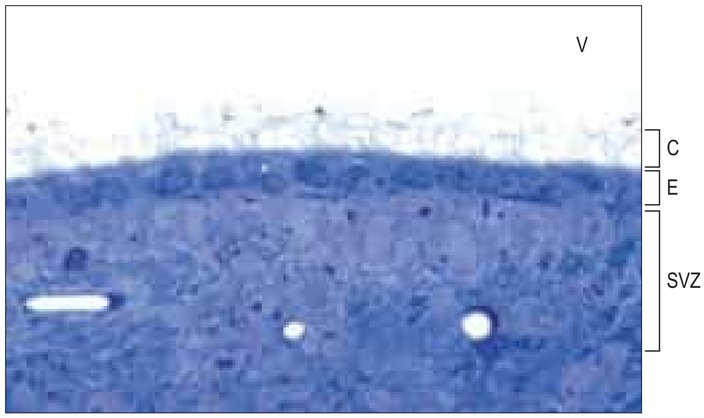

Fig. 3.17 Ciliated columnar epithelial lining of the lateral ventricle (V), overlying the subventricular zone (SVZ). C, cilia; E, ependymal cells. Mouse tissue, toluidine blue stained resin section.

ependymal lining of the ventricles but four major types have been described. These are: general ependymal, which overlies grey matter; general ependymal, which overlies white matter; specialized areas of ependyma in the third and fourth ventricles; and choroidal epithelium.

The ependymal cells overlying areas of grey matter are cuboidal. Each cell bears approximately 20 central apical cilia, surrounded by short microvilli. The cells are joined by gap junctions and desmosomes. Beneath them there may be a subependymal (or subventricular) zone, from two to three cells deep, consisting of cells that generally resemble ependymal cells. In rodents, the subventricular zone contains neural progenitor cells, which can give rise to new neurones, but the existence of these stem cells in the adult human brain is controversial (Sanai et al 2011, Kempermann 2011). The capillaries beneath the ependymal cells have no fenestrations and few transcytotic vesicles, which is typical of the CNS. Where the ependyma overlies myelinated tracts of white matter, the cells are much flatter and few are ciliated. There are gap junctions and desmosomes between these cells, but their lateral margins interdigitate, unlike their counterparts overlying grey matter. No subependymal zone is present.

Specialized areas of ependymal cells called the circumventricular organs are found in four areas around the margins of the third ventricle: namely, the lining of the median eminence of the hypothalamus; the subcommissural organ; the subfornical organ; and the vascular organ of the lamina terminalis. The area postrema, at the inferoposterior limit of the fourth ventricle, has a similar structure. In all of these sites the ependymal cells are only rarely ciliated and their ventricular surfaces bear many microvilli and apical blebs. They have numerous mitochondria, well-formed Golgi complexes and rather flattened basal nuclei. They are joined laterally by tight junctions, which form a barrier to the passage of materials across the ependyma, and by desmosomes. Many of the cells are tanycytes (ependymal astrocytes) and have basal processes that project into the perivascular space surrounding the underlying capillaries. Significantly, these capillaries are fenestrated and therefore do not form a blood–brain barrier. It is believed that neuropeptides can pass from nervous tissue into the CSF by active transport through the ependymal cells in these specialized areas, and so access a wide population of neurones via the permeable ependymal lining of the rest of the ventricle.

The ependyma is highly modified where it lies adjacent to the vascular layer of the choroid plexuses.

Choroid plexus

The choroid plexus forms the CSF and actively regulates the concentration of molecules in the CSF. It consists of highly vascularized masses of pia mater enclosed by pockets of ependymal cells. The ependymal cells resemble those of the circumventricular organs, except that they do not have basal processes, but form a cuboidal epithelium that rests on a basal lamina adjacent to the enclosed fold of meningeal pia mater and its capillaries (Fig. 3.18). The cells have numerous long microvilli with only a few cilia interspersed between them. They also have many mitochondria, large Golgi complexes and basal nuclei, features consistent with their secretory activity; they produce most components of the CSF. They are linked by tight junctions forming a transepithelial barrier (a component of the blood–CSF barrier), and by desmosomes. Their lateral margins are highly folded.

The choroid plexus has a villous structure where the stroma is composed of pial meningeal cells, and contains fine bundles of collagen and blood vessels. Choroidal capillaries are lined by a fenestrated endothelium. During fetal life, erythropoiesis occurs in the stroma, which is occupied by bone marrow-like cells. In adult life, the stroma contains phagocytic cells, which, together with the cells of the choroid plexus epithelium, phagocytose particles and proteins from the ventricular lumen.

Age-related changes occur in the choroid plexus, which can be detected by neuroimaging. Calcification of the choroid plexus can be detected by X-ray or CT scan very rarely in individuals in the first decade of life and in the majority in the eighth decade. The incidence of calcification rises sharply, from 35% of CT scans in the fifth decade to 75% in the sixth decade. Visible calcification is usually restricted to the glomus region of the choroid plexus, i.e. the vascular bulge in the choroid plexus as it curves to follow the anterior wall of the lateral ventricle into the temporal horn.

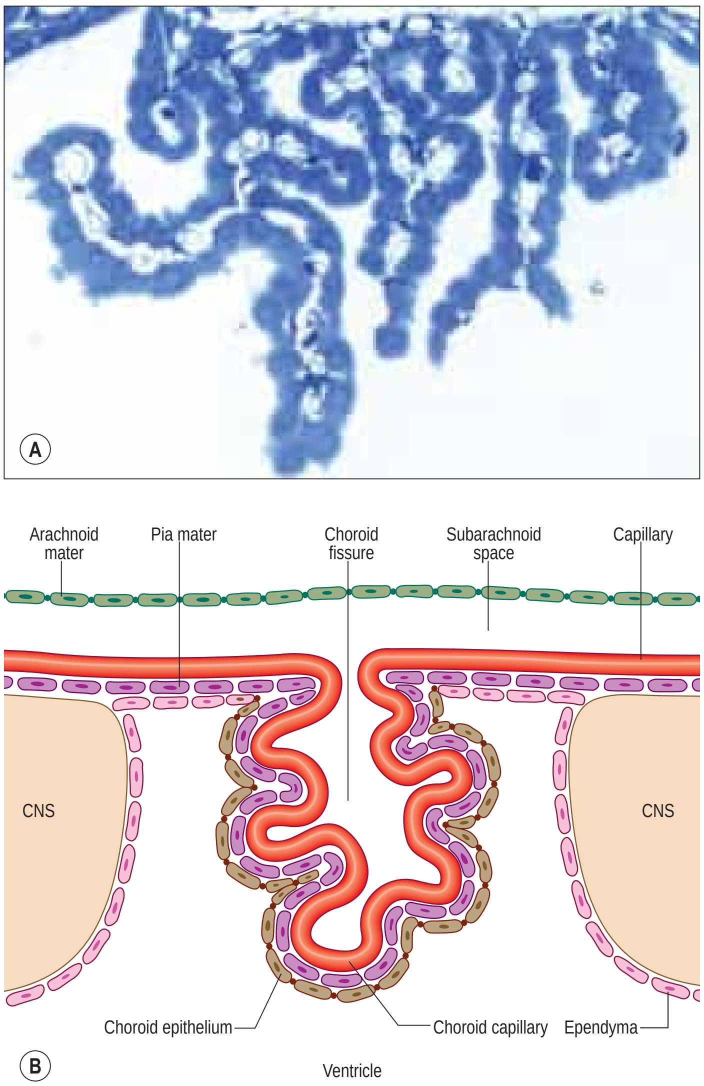

Fig. 3.18 **A**, A choroid plexus within the lateral ventricle. Frond-like projections of vascular stroma derived from the pial meninges are covered with a low columnar epithelium that secretes cerebrospinal fluid. Mouse tissue, toluidine blue stained resin section. **B**, The arrangement of tissues forming the choroid plexus.

MICROGLIA

Microglia are the endogenous immune cells of the brain (Kettenmann et al 2011, Eggen et al 2013). They originate from an embryonic monocyte precursor and invade the brain early during development. While the invading cells have an ameboid morphology, microglial cells in a mature brain are highly ramified cells. They have elongated nuclei, scant cytoplasm and several highly branched processes. They occupy a defined territory in the brain parenchyma and are found in all areas of the CNS including optic nerve, retina and spinal cord. Their density shows little variation.

Resting microglia, a term used to refer to microglia in the normal brain, should more accurately be described as surveying microglia. Microglial processes are fast-moving structures that rapidly scan their territory while the soma remains fixed in position. Microglial cells express receptors for neurotransmitters and thus can sense neuronal activity. It is likely that they interact with synapses, from which it has been inferred that they may influence synaptic transmission.

All pathological changes in the brain result in the activation of microglial cells (Fig. 3.19), e.g. activated microglia are found in the brain tissue of multiple sclerosis, Alzheimer's disease and stroke patients. The most common indication of their activation is a change from a ramified to an ameboid morphology, which may occur within a few hours of the onset of injury or disease process.

In general, microglia respond to two types of signal: 'on' signals, which either appear *de novo* or are strongly upregulated, e.g. cell wall components of invading bacteria; and 'off' signals, which are normally present but disappear or decrease in pathological states, e.g. defined cytokines or neurotransmitters. Both types of event are interpreted as signals for activation. The functional repertoire of activated microglia includes proliferation; migration to the site of injury; expression of major histocompatibility complex (MHC) II molecules to interact with infiltrating lymphocytes; and the release of a variety of different substances including chemokines, cytokines and growth factors. These cells are therefore capable of significantly influencing ongoing pathological processes.

Microglial cells are the professional phagocytes of the nervous system and actively migrate through tissue. A number of factors such as ATP and complement factors act as chemoattractants. This behaviour is relevant not only in pathology but also during development where microglial cells remove apoptotic cells. After a pathological insult, microglial cells revert to their surveying phenotype, re-acquiring a ramified morphology.

Entry of inflammatory cells into the brain

Although the CNS has long been considered to be an immunologically privileged site, lymphocyte and macrophage surveillance of the brain may be a normal, but very low-grade, activity that is enhanced in disease. Lymphocytes can enter the brain in response to virus infections and as part of the autoimmune response in multiple sclerosis. Activated, but not resting, lymphocytes pass through the endothelium of small venules, a process that requires the expression of recognition and adhesion molecules (induced following cytokine activation), and subsequently migrate into the brain parenchyma. Within the CNS, microglia can be induced by T-cell cytokines to act as efficient antigen-presenting cells. After leaving the CNS, lymphocytes probably drain along lymphatic pathways to regional cervical lymph nodes.

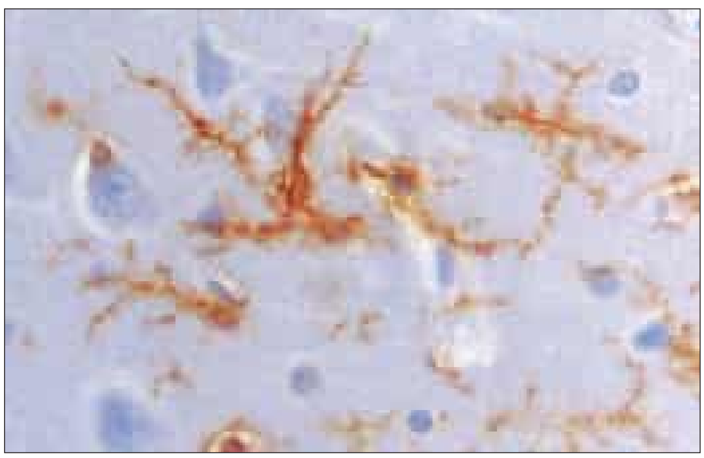

Fig. 3.19 Activated microglial cells in the human central nervous system, in a biopsy from a patient with Rasmussen's encephalitis, visualized using MHC class II antigen immunohistochemistry. (Courtesy of Dr Norman Gregson, Division of Neurology, GKT School of Medicine, London.)

Monocytes enter the CNS in the early stages of infarction and autoimmune disease and, in particular, in pyogenic infections, probably by passing through the endothelium of local vessels. Once in the brain, monocytes are difficult to distinguish from intrinsic microglia because both cell types express a similar marker profile. During the inflammatory phase of meningitis, polymorphonuclear leukocytes and lymphocytes pass into the CSF through the endothelium of large veins in the subarachnoid space. Recent developments in research on brain inflammatory disorders are reviewed in Anthony and Pitossi (2013).

PERIPHERAL NERVES

Afferent nerve fibres connect peripheral receptors to the CNS; they are derived from neuronal somata located either in special sense organs (e.g. the olfactory epithelium) or in the sensory ganglia of the craniospinal nerves. Efferent nerve fibres connect the CNS to the effector cells and tissues and are the peripheral axons of neurones with somata in the central grey matter.

Peripheral nerve fibres are grouped in widely variable numbers into bundles (fasciculi). The size, number and pattern of fasciculi vary in different nerves and at different levels along their paths (Fig. 3.20). Their number increases and their size decreases some distance proximal to a point of branching. Where nerves are subjected to pressure, e.g. deep to a retinaculum, fasciculi are increased in number but reduced in size, and the amount of associated connective tissue and degree of vascularity also increase. At these points, nerves may occasionally show a pink, fusiform dilation, sometimes termed a pseudoganglion or gangliform enlargement.

CLASSIFICATION OF PERIPHERAL NERVE FIBRES

Classification of peripheral nerve fibres is based on various parameters such as conduction velocity, function and fibre diameter. Of two classifications in common use, the first divides fibres into three major classes, designated A, B and C, corresponding to peaks in the distribution of their conduction velocities. In humans, this classification is used mainly for afferent fibres from the skin. Group A fibres are subdivided into α, β, γ and δ subgroups; fibre diameter and conduction velocity are proportional in most fibres. Group Aα fibres are the largest and conduct most rapidly, and C fibres are the smallest and slowest.

The largest afferent axons (Aα fibres) innervate encapsulated cutaneous mechanoreceptors, Golgi tendon organs and muscle spindles, and some large alimentary enteroceptors. Aβ fibres form secondary endings on some muscle spindle (intrafusal) fibres and also innervate cutaneous and joint capsule mechanoreceptors. Aδ fibres innervate thermoreceptors, stretch-sensitive free endings, hair receptors and nociceptors, including those in dental pulp, skin and connective tissue. Aγ fibres are exclusively fusimotor to plate and trail endings on intrafusal muscle fibres. B fibres are myelinated autonomic preganglionic efferent fibres. C fibres are unmyelinated and have thermoreceptive, nociceptive and interoceptive functions, including the perception of slow, burning pain and visceral pain. This scheme can be applied to fibres of both spinal and cranial nerves except perhaps those of the olfactory nerve, where the fibres form a uniquely small and slow group. The largest somatic efferent fibres (Aα) innervate extrafusal muscle fibres (at motor endplates) exclusively and conduct at a maximum of 120 m/s. Fibres to fast twitch muscles are larger than those to slow twitch muscle. Smaller (Aγ) fibres of gamma motor neurones, and autonomic preganglionic (B) and postganglionic (C) efferent fibres conduct, in order, progressively more slowly (40 m/s to less than 10 m/s).

A different classification, used for afferent fibres from muscles, divides fibres into groups I–IV on the basis of their calibre; groups I–III are myelinated and group IV is unmyelinated. Group I fibres are large (12–22 μm), and include primary sensory fibres of muscle spindles (group Ia) and smaller fibres of Golgi tendon organs (group Ib). Group II fibres are the secondary sensory terminals of muscle spindles, with diameters of 6–12 μm. Group III fibres, 1–6 μm in diameter, have free sensory endings in the connective tissue sheaths around and within muscles and are nociceptive and, in skin, also thermosensitive. Group IV fibres are unmyelinated, with diameters below 1.5 μm; they include free endings in skin and muscle, and are primarily nociceptive.

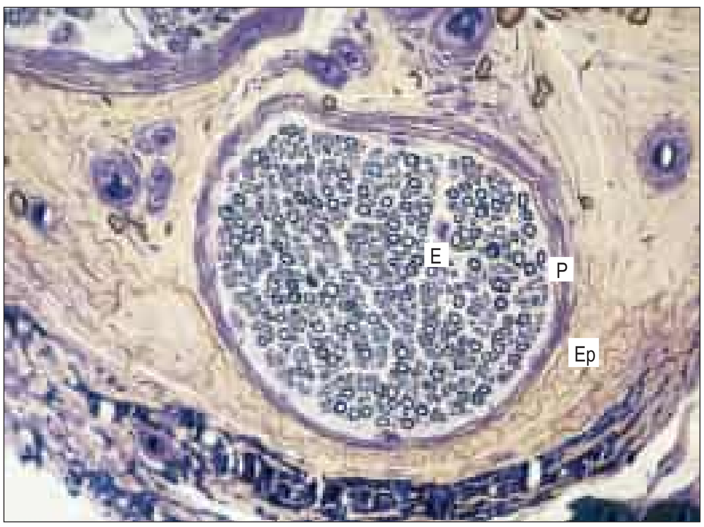

Fig. 3.20 A transverse section of a biopsied human sural nerve, showing the arrangement of the connective tissue sheaths. Individual axons, myelinated and unmyelinated, are arranged in a small fascicle bounded by a perineurium. Abbreviations: P, perineurium; Ep, epineurium; E, endoneurium. (Courtesy of Professor Susan Standring, GKT School of Medicine, London.)

CONNECTIVE TISSUE SHEATHS

Nerve trunks, whether uni- or multifascicular, are limited externally by an epineurium, which is connected to surrounding tissues by mesoneurium. Mesoneurium is a loose connective tissue sheath (see Ch. 2) containing the extrinsic, segmental blood supply of the nerve, and so is of clinical importance in nerve injury. Individual fasciculi of the nerve trunk are enclosed by a multilayered perineurium, which in turn surrounds the endoneurium or intrafascicular connective tissue (see Fig. 3.20).

Epineurium

Epineurium is a condensation of loose (areolar) connective tissue derived from mesoderm. As a general rule, the more fasciculi present in a peripheral nerve, the thicker the epineurium. Epineurium contains fibroblasts, collagen (types I and III) and variable amounts of fat, and it cushions the nerve it surrounds. Loss of this protective layer may be associated with pressure palsies seen in wasted, bedridden patients. The epineurium also contains lymphatics (which probably pass to regional lymph nodes) and blood vessels, vasa nervorum, that pass across the perineurium to communicate with a network of fine vessels within the endoneurium, forming the intrinsic system of vascular plexuses.

Perineurium

Perineurium extends from the CNS–PNS transition zone to the periphery, where it is continuous with the capsules of muscle spindles and encapsulated sensory endings, but ends openly at unencapsulated endings and neuromuscular junctions. It consists of alternating layers of flattened polygonal cells (thought to be derived from fibroblasts) and collagen. It can often contain 15–20 layers of such cells, each layer enclosed by a basal lamina up to 0.5 μm thick. Within each layer the cells interdigitate along extensive tight junctions; their cytoplasm typically contains vesicles and bundles of microfilaments and their plasma membrane often shows evidence of pinocytosis. These features are consistent with the function of the perineurium as a metabolically active diffusion barrier; together with the blood–nerve barrier, the perineurium is thought to play an essential role in maintaining the osmotic milieu and fluid pressure within the endoneurium. Lymphatic vessels have not been detected in the perineurium.

Endoneurium

Strictly speaking, the term endoneurium is restricted to intrafascicular connective tissue and excludes the perineurial partitions within fascicles. Endoneurium consists of a fibrous matrix composed predominantly of type III collagen (reticulin) fibres, characteristically organized in fine bundles lying parallel to the long axis of the nerve, and condensed around individual Schwann cell–axon units and endoneurial vessels. The fibrous and cellular components of the endoneurium are bathed in endoneurial fluid at a slightly higher pressure than that outside in the surrounding epineurium. The major cellular constituents of the endoneurium are Schwann cells and endothelial cells; minor components are fibroblasts (constituting approximately 4% of the total endoneurial cell population), resident macrophages and mast cells. Schwann cell–axon units and blood vessels are enclosed within individual basal laminae and therefore isolated from the other cellular and extracellular components of the endoneurium.

Endoneurial arterioles have a poorly developed smooth muscle layer and do not autoregulate well. In sharp contrast, epineurial and perineurial vessels have a dense perivascular plexus of peptidergic, serotoninergic and adrenergic nerves. There are free nerve endings in all layers of neural connective tissue sheaths and there are some encapsulated (Pacinian) corpuscles in the endoneurium. These probably contribute to the acute sensitivity of nerves trapped in fibrosis after injury or surgery.

SCHWANN CELLS

Schwann cells are the major glial type in the PNS. *In vitro* they are fusiform in appearance. Both *in vitro* and *in vivo*, Schwann cells ensheathe peripheral axons, and myelinate those greater than 2 µm in diameter. In a mature peripheral nerve, they are distributed along the axons in longitudinal chains; the geometry of their association depends on whether the axon is myelinated or unmyelinated. In myelinated axons the territory of a Schwann cell defines an internode.

The molecular phenotype of mature myelin-forming Schwann cells is different from that of mature non-myelin-forming Schwann cells. Adult myelin-forming Schwann cells are characterized by the presence of several myelin proteins, some, but not all, of which are shared with oligodendrocytes and central myelin. In contrast, expression of the low-affinity neurotrophin receptor ($p75^{NTR}$) and GFAP intermediate filament protein (which differs from the CNS form in its post-translational modification) characterizes adult non-myelin-forming Schwann cells.

Schwann cells arise from Schwann cell precursors that, in turn, are generated from multipotent cells of the neural crest. Neuronal signals regulate many aspects of Schwann cell behaviour in developing and postnatal nerves. Axon-associated signals appear to control the proliferation of developing Schwann cells and their precursors; the developmentally programmed death of those precursors in order to match numbers of axons and glia within each peripheral nerve bundle; the production of basal laminae by Schwann cells; and the induction and maintenance of myelination. Axonal neuregulin 1 signalling via ErbB2/B3 receptors on Schwann cells is essential for Schwann cell myelination and also determines myelin thickness. An extensive literature supports the view that Schwann cells are key players in the acute injury response in the PNS (see Commentary 1.6), helping to provide a microenvironment that facilitates axonal regrowth (Birch 2011). Few Schwann cells persist in chronically denervated nerves. For further reading about Schwann cells, see Kidd et al (2013).

Unmyelinated axons

Unmyelinated axons are commonly 1.0 µm in diameter, although some may be 1.5 µm or even 2 µm in diameter. Groups of up to 10 or more small axons (0.15–2.0 µm in diameter) are enclosed within a chain of overlapping Schwann cells that is surrounded by a basal lamina. Within each Schwann cell, individual axons are usually sequestered from their neighbours by delicate processes of cytoplasm. It seems likely, on the basis of quantitative studies in subhuman primates, that axons from adjacent cord segments may share Schwann cell columns; this phenomenon may play a role in the evolution of neuropathic pain after nerve injury. In the absence of a myelin sheath and nodes of Ranvier, action potential propagation along unmyelinated axons is not saltatory but continuous, and relatively slow (0.5–4.0 m/s).

Myelinated axons

Myelinated axons (**Fig. 3.21**) have a 1 : 1 relationship with their ensheathing Schwann cells. The territorial extent of individual Schwann cells varies directly with the diameter of the axon they surround, from 150 to 1500 µm. Specialized domains of axo-glial interaction define nodes of Ranvier and their neighbouring compartments, paranodes and juxtaparanodes (Pereira et al 2012; **Fig. 3.22**). These domains contain multiprotein complexes characterized by unique sets of transmembrane and cytoskeletal proteins and clusters of ion channels; the mechanisms regulating channel clustering and node formation remain a subject of intense scrutiny (Peles and Salzer 2000, Poliak and Peles 2003, Horresh et al 2008). The region under the compact myelin sheath that extends between two juxtaparanodes is the internode. The molecular domains of myelinated axons, including that of the axon initial segment are reviewed in Buttermore et al (2013)).

Fig. 3.21 An electron micrograph of a transverse section of biopsied human sural nerve, showing a myelinated axon and several unmyelinated axons (A), ensheathed by Schwann cells (S). (Courtesy of Professor Susan Standring, GKT School of Medicine, London.)

Schwann cell cytoplasm forms a continuous layer only in the perinuclear (mid-internodal) and paranodal regions, where it forms a collar from which microvilli project into the nodal gap substance. Elsewhere it is dispersed as a lace-like network over the inner (adaxonal) and outer (abaxonal) surfaces of the myelin sheath.

Nodes of Ranvier

The nodal compartment consists of a short length of exposed axolemma, typically 0.8–1.1 µm long, surrounded by a nodal gap substance composed of various extracellular components, some of which may possess nerve growth-repulsive characteristics. Multiple processes (microvilli) protrude into the gap substance from the outer collar of Schwann cytoplasm and contact the nodal axolemma. Voltage-gated Na^+ channels, the cell adhesion molecules NrCAM and neurofascin-186, the cytoskeletal adaptor ankyrin G25,26 and the actin-binding protein spectrin βIV are clustered at nodes. The calibre of the nodal axon is usually significantly less than that of the internodal axon, particularly in large-calibre fibres.

Paranodes

The axolemma on either side of a node is contacted by paranodal loops of Schwann cell cytoplasm via specialized septate junctions that spiral around the axon. The junctions are thought to form a partial diffusion barrier into the peri-axonal space; restrict the movement of K^+ channels from under the compact myelin; and limit lateral diffusion of membrane components. Caspr, contactin and their putative ligand NF155 (an isoform of neurofascin) are concentrated in paranodes.

Juxtaparanodes

The region of the axon lying just beyond the innermost paranodal junction is now recognized as a distinct domain defined by the localization of voltage-gated K^+ channels (delayed-rectifier K^+ channels Kv1.1, Kv1.2 and their Kvb2 subunit). Clustering of Kv1 channels at the juxtaparanodal region depends on their association with the Caspr2/TAG-1 adhesion complex.

Schmidt–Lanterman incisures

Schmidt–Lanterman incisures are helical decompactions of internodal myelin that appear as funnel-like profiles in teased fibre preparations labelled for markers of non-compacted myelin (e.g. MAG, Cx32). At an incisure the major dense line of the myelin sheath splits to enclose a continuous spiral band of cytoplasm passing between abaxonal and adaxonal layers of Schwann cell cytoplasm. The minor dense line of incisural myelin is also split, creating a channel connecting the peri-axonal space with the endoneurial extracellular fluid. The function of

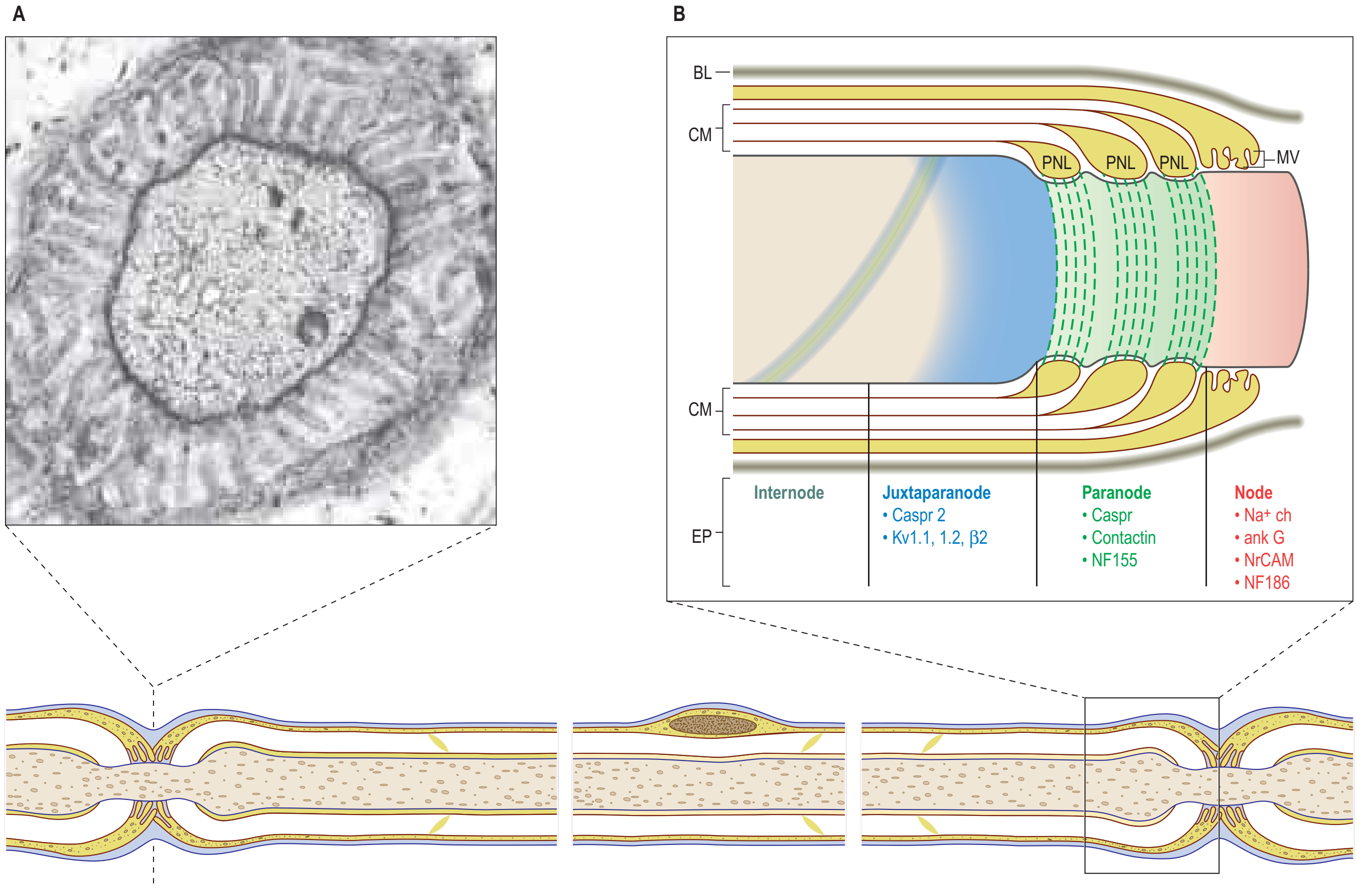

Fig. 3.22 The general plan of a peripheral myelinated nerve fibre in longitudinal section, including one complete internodal segment and two adjacent paranodal bulbs, used as a key for the more detailed microarchitecture of specific subregions. **A**, A transverse electron microscope section through the centre of a node of Ranvier, with numerous finger-like processes of adjacent Schwann cells converging towards the nodal axolemma. Many microtubules and neurofilaments are visible within the axoplasm. **B**, The arrangement of the axon, myelin sheath and Schwann cell cytoplasm at the node of Ranvier, in the paranodal bulbs and in the juxtaparanodal region. The axon is myelinated by a Schwann cell surrounded by a basal lamina (BL). Only a portion of the internode, which is located beneath the compact myelin (CM) sheath, is shown. A spiral of paranodal (green) and juxtaparanodal (blue) proteins extends into the internode; this spiral is apposed to the inner mesaxon of the myelin sheath (not shown). K^+ channels and Caspr2 are concentrated in the juxtaparanodal region. In the paranodal region, the compact myelin sheath opens up into a series of paranodal cytoplasmic loops (PNL) that invaginate and closely appose the axon, forming a series of septum-like junctions that spiral around the axon. Caspr, contactin and an isoform of neurofascin (NF155) are concentrated in this region. At the node, numerous microvilli (MV) project from the outer collar of the Schwann cell to contact the axolemma. The axon is enormously enriched in intramembranous particles at the node that correspond to Na^+ channels (Na^+ ch). Ankyrin G (ank G) isoforms and the cell adhesion molecules NrCAM and NF186 are also concentrated in this region. (A, Courtesy of Professor Susan Standring, GKT School of Medicine, London. B, Redrawn from Peles and Salzer 2000.)

incisures is not known; their structure suggests that they may participate in transport of molecules across the myelin sheath.

SATELLITE CELLS

Many non-neuronal cells of the nervous system have been called satellite cells, including small, round extracapsular cells in peripheral ganglia, ganglionic capsular cells, Schwann cells, any cell that is closely associated with neuronal somata, and precursor cells associated with striated muscle fibres (Hanani 2010). Within the nervous system, the term is most commonly reserved for flat, epithelioid cells (ganglionic glial cells, capsular cells) that surround the neuronal somata of peripheral ganglia (see Fig. 3.23). Their cytoplasm resembles that of Schwann cells, and their deep surfaces interdigitate with reciprocal infoldings in the membranes of the enclosed neurones.

Enteric glia

Enteric nerves lack an endoneurium and so do not have the collagenous coats of other peripheral nerves. The enteric ganglionic neurones are supported by glia that closely resemble astrocytes; they contain more GFAP than non-myelinating Schwann cells and do not produce a basal lamina. Evidence for their roles in gut function is reviewed in Gulbransen and Sharkey (2012).

Olfactory ensheathing glia

The olfactory system is unusual because it supports neurogenesis throughout life. Olfactory receptor neurones are continuously renewed from horizontal basal stem cells in the olfactory epithelium (Leung et al 2007, Forni and Wray 2012). They extend new axons through the lamina propria and cribriform plate into the CNS environment of the olfactory bulb, where they synapse with second-order neurones. Olfactory ensheathing cells (OECs, also known as olfactory ensheathing glia) accompany olfactory axons from the lamina propria of the olfactory epithelium to their synaptic contacts in the glomeruli of the olfactory bulbs and are thought to play a role in directing them to their correct position in the olfactory bulb (Higginson and Barnett 2011). This unusual arrangement is unique; elsewhere in the nervous system the territories of peripheral and central glia are clearly demarcated at CNS–PNS transition zones. OECs and the end-feet of astrocytes lying between the bundles of olfactory axons both contribute to the glia limitans at the pial surface of the olfactory bulbs.

OECs share many properties with Schwann cells and express similar antigenic and morphological properties. They ensheathe olfactory sensory axons in a manner comparable to the relationship that exists transitorily between Schwann cells and axons in very immature peripheral nerves, i.e. they surround, but do not segregate, bundles of up to 50 fine unmyelinated axons to form approximately 20 fila olfactoria. Both OECs and Schwann cells can myelinate axons, even though normally none of the axons in the olfactory nerve is myelinated. It was thought that OECs shared a common origin with olfactory receptor neurones in the olfactory placode, but recent fate-mapping experiments in chicken embryos and genetic linkage-tracing studies in mice have shown that OECs are derived from neural crest cells (Forni and Wray 2012).

OECs have a malleable phenotype. There may be several subtypes: some OECs express GFAP as either fine filaments or more diffusely in their cytoplasm, and some express $p75^{NTR}$ and the O4 antigen.

BLOOD SUPPLY OF PERIPHERAL NERVES

The blood vessels supplying a nerve, end in a capillary plexus that pierces the perineurium. The branches of the plexus run parallel with the fibres, connected by short transverse vessels, forming narrow, rectangular meshes similar to those found in muscle. The blood supply of peripheral nerves is unusual. Endoneurial capillaries have atypically large diameters and intercapillary distances are greater than in many other tissues. Peripheral nerves have two separate, functionally independent vascular systems: an extrinsic system (regional nutritive vessels and epineurial vessels) and an intrinsic system (longitudinally running microvessels in the endoneurium). Anastomoses between the two systems produce considerable overlap between the territories of the segmental arteries. This unique pattern of vessels, together with a high basal nerve blood flow relative to metabolic requirements, means that peripheral nerves possess a high degree of resistance to ischaemia.

Blood–nerve barrier

Just as the neuropil within the CNS is protected by a blood–brain barrier, the endoneurial contents of peripheral nerve fibres are protected by a blood–nerve barrier and by the cells of the perineurium. The blood–nerve barrier operates at the level of the endoneurial capillary walls, where the endothelial cells are joined by tight junctions, and are non-fenestrated and surrounded by continuous basal laminae. The barrier is much less efficient in dorsal root ganglia and autonomic ganglia and in the distal parts of peripheral nerves.

GANGLIA

Ganglia are aggregations of neuronal somata and are of varying form and size. They occur in the dorsal roots of spinal nerves; in the sensory roots of the trigeminal, facial, vestibulocochlear, glossopharyngeal and vagal cranial nerves; and in the peripheral autonomic nervous system (ANS). Each ganglion is enclosed within a capsule of fibrous connective tissue and contains neuronal somata and neuronal processes. Enteric ganglia are an exception to this rule; they resemble the CNS in both structure and function, and are not covered by a connective tissue capsule. Some ganglia, particularly in the ANS, contain axons that originate from neuronal somata that lie elsewhere in the nervous system and which pass through the ganglia without synapsing.

Sensory ganglia

The sensory ganglia of dorsal spinal roots (Fig. 3.23) and the ganglia of the trigeminal, facial, glossopharyngeal and vagal cranial nerves are enclosed in a periganglionic connective tissue capsule that resembles the perineurium surrounding peripheral nerves. Ganglionic neurones are unipolar (sometimes called pseudounipolar, see above). They have spherical or oval somata of varying size, aggregated in groups between fasciculi of myelinated and unmyelinated nerve fibres. For each neurone, a single axodendritic process bifurcates into central and peripheral processes; in myelinated fibres the junction occurs at a node of Ranvier. The peripheral process terminates in a sensory ending and, because it conducts impulses towards the soma, it functions as an elongated dendrite, strictly speaking. However, it has the typical structural and functional properties of a peripheral axon, and is conventionally described as an axon.

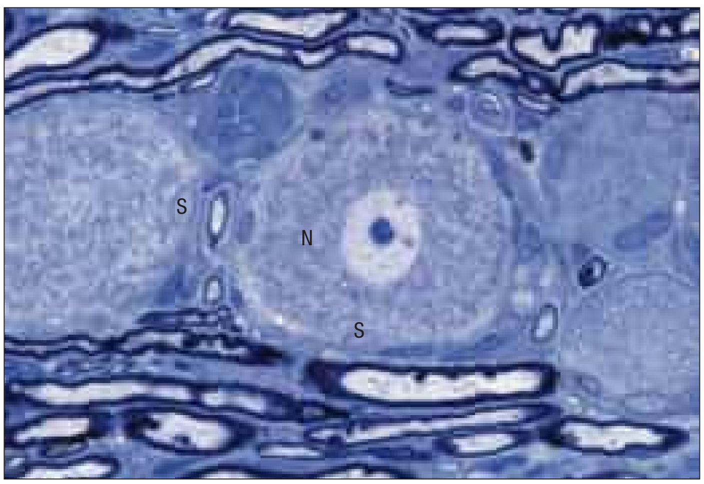

Fig. 3.23 Sensory neurones in a dorsal root ganglion (rat). Neurones (N) are typically variable in size but all are encapsulated by satellite cells (S). Myelinated axons are seen above and below the neuronal somata. Toluidine blue stained resin section. (Courtesy of Dr Clare Farmer, King's College, London.)

Each neuronal soma is surrounded by a sheath of satellite glial cells (SGCs). (A notable exception is the spiral, or cochlear, ganglion, where most neuronal somata are myelinated, presumably contributing to fast electrical transmission.) The axodendritic process and its peripheral and central divisions, ensheathed by Schwann cells, lie outside the SGC sheath. All the cells in the ganglion lie within a highly vascularized connective tissue that is continuous with the endoneurium of the nerve root. In dorsal root ganglia there is no clear regional mapping of the innervated body regions. In contrast, each of the three nerve branches (ophthalmic, maxillary and mandibular) of the trigeminal nerve is mapped to a different part of the trigeminal ganglion. Although sensory neurones receive no synapses, they are endowed with receptors for numerous neurotransmitters and hormones, and can thus communicate chemically amongst themselves and with SGCs.

SGCs are the main type of glial cell in sensory ganglia. They share several properties with astrocytes, including expression of glutamine synthetase and various neurotransmitter transporters. In addition, like astrocytes, the SGCs that surround a neurone are coupled by gap junctions and express receptors for ATP. Unlike astrocytes, SGCs completely surround individual sensory neurones (and more rarely two or three neurons) in a glial sheath. They undergo major changes as a result of injury to peripheral nerves, and appear to contribute to chronic pain in a number of animal pain models.

Herpes zoster Primary infection with the varicella zoster virus causes chickenpox. Following recovery, the virus remains dormant within dorsal root ganglia or trigeminal ganglia, mostly in the neurones, and less commonly in the SGCs. Reactivation of the virus leads to herpes zoster (shingles), which involves the dermatome(s) supplied by the affected sensory nerve(s). Diagnostic signs are severe pain, erythema and blistering as occurs in chickenpox, often confined to one of the divisions of the trigeminal nerve or to a spinal nerve dermatome. Herpes zoster involving the geniculate ganglion compresses the facial nerve and results in a lower motor neurone facial paralysis, known as Ramsay Hunt syndrome. Occasionally, if the vestibulocochlear nerve becomes involved, there is vertigo, tinnitus and some deafness. The most important complication of herpes zoster is post-herpetic neuralgia, a severe and persistent pain that is highly refractory to treatment.

Autonomic ganglia

The main types of cell in autonomic ganglia are the ganglionic neurones, small intensely fluorescent (SIF) cells and satellite glial cells (SGCs). Most of the neurones have somata ranging from 25 to 50 μm and complex dendritic fields; dendritic glomeruli have been observed in ganglia in experimental animals. Ganglionic neurones receive many axodendritic synapses from preganglionic axons; axosomatic synapses are less numerous. Postganglionic fibres commonly arise from the initial stem of a large dendrite and produce few or no collateral processes. Given their close relationship to the ganglionic neurones, autonomic SGCs may have the potential to influence synaptic transmission. SIF cells are characterized by being smaller than the neurones and by having numerous granules that contain noradrenaline (norepinephrine), dopamine and serotonin. They are almost completely invested by a sheath of SGCs and receive and make synapses; their physiological role is currently obscure, but they lend credence to the idea that autonomic ganglia are far more than simple relay stations.

Sympathetic neurones are multipolar and their dendritic trees, on which preganglionic motor axons synapse, are more elaborate than those of parasympathetic neurones (Fig. 3.24). The neurones are surrounded by a mixed neuropil of afferent and efferent fibres, dendrites, synapses and non-neural cells. There is considerable variation in the ratio of pre- and postganglionic fibres in different types of ganglion. Preganglionic sympathetic axons may synapse with many postganglionic neurones for the wide dissemination and perhaps amplification of sympathetic activity, a feature not found to the same degree in parasympathetic ganglia. Dissemination may also be achieved by connections with ganglionic interneurones or by the diffusion within the ganglion of transmitter substances produced either locally (paracrine effect) or elsewhere (endocrine effect). Some axons within a ganglion may be efferent fibres en route to another ganglion, or afferents from viscera and glands. These fibres may synapse with neurones in the

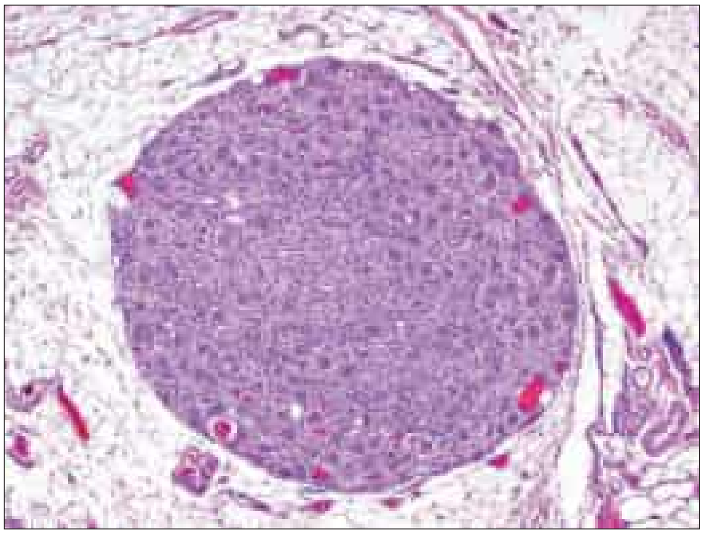

Fig. 3.24 A parasympathetic autonomic ganglion from a human stomach. Large neuronal somata, some with nuclei and prominent nucleoli in the plane of section, are encapsulated by satellite cells and surrounded by nerve fibres and non-neuronal cells. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

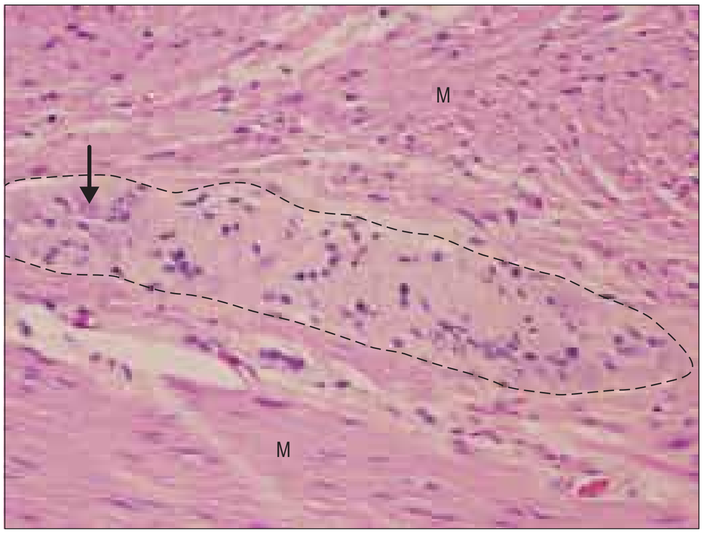

Fig. 3.25 An enteric ganglion (outlined) of the myenteric (Auerbach's) plexus between the inner circular and outer longitudinal layers of smooth muscle (M) in the wall of the human intestine. An enteric ganglionic neurone is arrowed. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

ganglion, e.g. substance P-containing axons of dorsal root neurones synapse on neurones in prevertebral ganglia, thereby enabling interactions between the sensory system and the ANS.

Enteric ganglia

The enteric nervous system (ENS) lies within the walls of the gastrointestinal tract (see Fig. 2.15 for the layers of a typical viscus) and includes the myenteric and submucosal plexuses and associated ganglia (Furness 2012, Neunlist et al 2013). The ganglionic neurones (**Fig. 3.25**) serve different functions, including the regulation of gut motility (in conjunction with interstitial cells of Cajal (Huizinga et al 2009)), mucosal transport and mucosal blood flow. Unlike the other two divisions of the ANS, the ENS is largely independent of the CNS, and the extrinsic autonomic fibres that supply the gut wall exert only modulatory effects on it. Submucosal neurones, together with sympathetic axons, regulate the local blood flow.

Hirschsprung's disease is a congenital disease in which dysfunctional neural crest migration means that the ganglia of both the myenteric and submucosal plexuses in the distal bowel fail to develop. The resulting lack of propulsive activity in the aganglionic bowel leads to functional obstruction and megacolon, which can be life-threatening. Around 1 in 5,000 infants is born with the condition and is typically diagnosed in the neonatal period. Treatment usually consists of removing the diseased intestinal segment.

The enteric plexuses consist of sensory neurones, interneurones and a variety of motor neurones. These neurones are endowed with receptors for a large number of neurotransmitters and also release a variety of neurotransmitters. All classes of enteric neurone are equally distributed along the entire ganglionic network; consequently the ENS consists of numerous repeating modules.

The myenteric plexus contains the motor neurones that control the movements of gastrointestinal smooth muscle. The main excitatory neurotransmitter is acetylcholine, which may be co-localized with an excitatory peptide (usually a tachykinin, such as substance P). The main inhibitory neurotransmitter is nitric oxide (NO), released from neurones that may also release the inhibitory peptide vasoactive intestinal peptide (VIP).

An important function of myenteric neurones is to mediate the peristaltic reflex, which is induced by intestinal wall distension or by mechanical stimulation of the mucosa. These stimuli initiate contraction oral to the site of the stimulus, and relaxation anal to the site, creating a pressure gradient that propels the intestinal contents. Interstitial cells of Cajal (ICC) are pacemaker cells believed to integrate neuronal signals with rhythmic oscillations of muscle contraction; disturbance of ICC function may be a factor in a number of gastrointestinal disorders (Huizinga et al 2009).

Enteric glia are the main type of glial cell in the ENS. In some respects they resemble astrocytes, e.g. they form end-feet with blood vessels, respond to numerous chemical mediators, and are extensively coupled among themselves by gap junctions. They appear to play an important role in neuroprotection and in maintaining the integrity of the intestinal mucosal barrier.

DISPERSED NEUROENDOCRINE SYSTEM

Although the nervous, neuroendocrine and endocrine systems all operate by intercellular communication, they differ in the mode, speed and degree or localization of the effects produced (Day and Salzet 2002). The autonomic nervous system uses impulse conduction and neurotransmitter release to transmit information, and the responses induced are rapid and localized. The dispersed neuroendocrine system uses only secretion. It is slower and the induced responses are less localized, because the secretions, e.g. neuromediators, can act either on contiguous cells, or on groups of nearby cells reached by diffusion, or on distant cells via the blood stream. Many of its effector molecules operate in both the nervous system and the neuroendocrine system. The endocrine system proper, which consists of clusters of cells and discrete, ductless, hormone-producing glands, is even slower and less localized, although its effects are specific and often prolonged. These regulatory systems overlap in function, and can be considered as a single neuroendocrine regulator of the metabolic activities and internal environment of the organism, acting to provide conditions in which it can function successfully. Neural and neuroendocrine axes appear to cooperate to modulate some forms of immunological reaction; the extensive system of vessels, circulating hormones and nerve fibres that link the brain with all viscera are thought to constitute a neuroimmune network (**Fig. 3.26**).

Some cells can take up and decarboxylate amine precursor compounds (amine precursor uptake and decarboxylation, or APUD, cells). They are characterized by dense-core cytoplasmic granules (see Fig. 2.6), similar to the neurotransmitter vesicles seen in some types of neuronal terminal. The group includes cells described as chromaffin cells (phaeochromocytes), derived from neuroectoderm and innervated by preganglionic sympathetic nerve fibres. Chromaffin cells synthesize and secrete catecholamines (dopamine, noradrenaline (norepinephrine) or adrenaline (epinephrine)). Their name refers to the finding that their cytoplasmic store of catecholamines is sufficiently concentrated to give an intense yellow–brown colouration, the positive chromaffin reaction, when they are treated with aqueous solutions of chromium salts, particularly potassium dichromate. Classic chromaffin cells include clusters of cells in the suprarenal medulla; the para-aortic bodies, which secrete noradrenaline; paraganglia; certain cells in the carotid bodies; and small groups of cells irregularly dispersed among the paravertebral sympathetic ganglia, splanchnic nerves and prevertebral autonomic plexuses.

The alimentary tract contains a large population of cells of a similar type (previously called neuroendocrine or enterochromaffin cells) in its wall. These cells act as sensory transducers, activating intrinsic and extrinsic primary afferent neurones via their release of 5-hydroxytryptamine (5-HT, serotonin). The neonatal respiratory tract

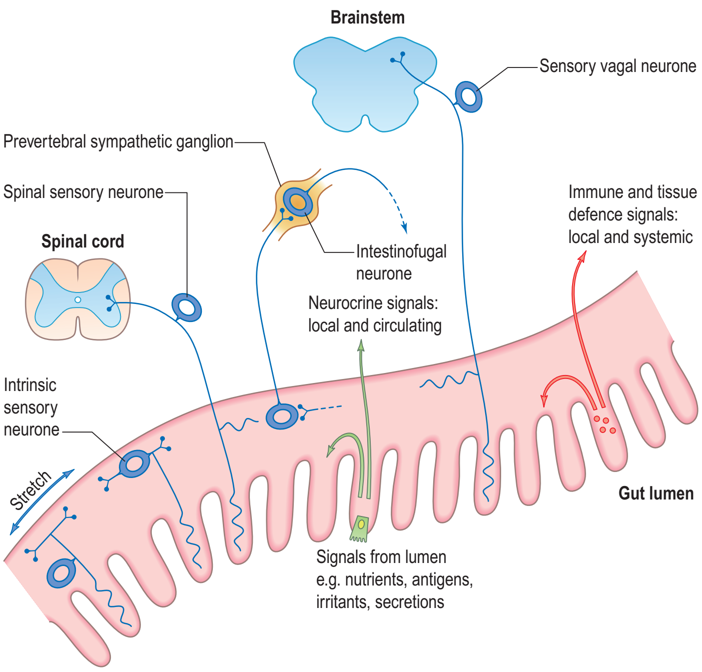

Fig. 3.26 The ways in which the nervous system, neuroendocrine system and immune system are integrated, demonstrated in the intestine. Neurocrine signals from enteric neuroendocrine cells and signals from immune defence cells (e.g. lymphocytes, macrophages and mast cells) act on other cells of those systems and on neurones with sensory endings in the intestinal wall, either locally or at a distance. Some neuronal soma lie within enteric ganglia in the gut wall; others have their bodies in peripheral ganglia. Neuronal signals may act locally, be transmitted to the CNS or enter a reflex pathway via sympathetic ganglia.

contains a prominent system of neuroendocrine cells, both dispersed and aggregated (neuroepithelial bodies); the numbers of both types decline during childhood. Merkel cells (see Commentary 1.3) in the basal epidermis of the skin store neuropeptides, which they release to associated nerve endings or other cells in a neuroendocrine role, in response to pressure and possibly other stimuli (Lucarz and Brand 2007). Experimental animal studies have revealed 5-HT-containing intraepithelial paraneurones in the urothelial lining of the urethra; these cells are thought to relay information from the luminal surface of the urethra to underlying sensory nerves.

A number of descriptions and terms have been applied to cells of this system in the older literature (see online text for details).

For further reading, see Day and Salzet (2002).

SENSORY ENDINGS

GENERAL FEATURES OF SENSORY RECEPTORS

There are three major forms of sensory receptor: neuroepithelial, epithelial and neuronal (**Fig. 3.27**).

A neuroepithelial receptor is a neurone with a soma lying near a sensory surface and an axon that conveys sensory signals into the CNS to synapse on second-order neurones. This is an evolutionarily primitive arrangement, and the only examples remaining in humans are the sensory neurones of the olfactory epithelium.

An epithelial receptor is a cell that is modified from a non-nervous sensory epithelium and innervated by a primary sensory neurone with a soma lying near the CNS, e.g. auditory receptors and taste buds. When activated, this type of receptor excites its neurone by neurotransmission across a synaptic gap.

A neuronal receptor is a primary sensory neurone that has a soma in a craniospinal ganglion and a peripheral axon ending in a sensory terminal. All cutaneous sensors and proprioceptors are of this type; their sensory terminals may be encapsulated or linked to special mesodermal or ectodermal structures to form a part of the sensory apparatus. The extraneural cells are not necessarily excitable, but create an appropriate environment for the excitation of the neuronal process.

The receptor stimulus is transduced into a graded change of electrical potential at the receptor surface (receptor potential), and this initiates an all-or-none action potential that is transmitted to the CNS. This may occur either in the receptor, when this is a neurone, or partly in the receptor and partly in the neurone that innervates it, in the case of epithelial receptors. Transduction varies with the modality of the stimulus, and usually causes depolarization of the receptor membrane (or hyperpolarization, in the retina). In mechanoreceptors, transduction may involve the deformation of membrane structure, which causes either strain or stretch-sensitive ion channels to open. In chemoreceptors, receptor action may resemble that for ACh at neuromuscular junctions. Visual receptors share similarities with chemoreceptors: light causes changes in receptor proteins, which activate G proteins, resulting in the release of second messengers and altered membrane permeability.

The quantitative responses of sensory endings to stimuli vary greatly and increase the flexibility of the functional design of sensory systems. Although increased excitation with increasing stimulus level is a common pattern ('on' response), some receptors respond to decreased stimulation ('off' response). Even unstimulated receptors show varying degrees of spontaneous background activity against which an increase or decrease in activity occurs with changing levels of stimulus. In all receptors studied, when stimulation is maintained at a steady level, there is an initial burst (the dynamic phase) followed by a gradual adaptation to steady level (the static phase). Though all receptors show these two phases, one or other may predominate, providing a distinction between rapidly adapting endings that accurately record the rate of stimulus onset, and slowly adapting endings that signal the constant amplitude of a stimulus, e.g. position sense. Dynamic and static phases are reflected in the amplitude and duration of the receptor potential and also in the frequency of action potentials in the sensory fibres. The stimulus strength necessary to elicit a response in a receptor, i.e. its threshold level, varies greatly between receptors, and provides an extra level of information about stimulus strength.

For further information on sensory receptors, see Nolte (2008).

FUNCTIONAL CLASSIFICATION OF RECEPTORS

Receptors have been classified in several ways. They may be classified by the modalities to which they are sensitive, such as mechanoreceptors (which are responsive to deformation, e.g. touch, pressure, sound waves, etc.), chemoreceptors, photoreceptors and thermoreceptors. Some receptors are polymodal, i.e. they respond selectively to more than one modality; they usually have high thresholds and respond to damaging stimuli associated with irritation or pain (nociceptors).

Another widely used classification divides receptors on the basis of their distribution in the body into exteroceptors, proprioceptors and interoceptors. Exteroceptors and proprioceptors are receptors of the somatic afferent components of the nervous system, while interoceptors are receptors of the visceral afferent pathways.

Exteroceptors respond to external stimuli and are found at, or close to, body surfaces. They can be subdivided into the general or cutaneous sense organs and special sensory organs. General sensory receptors include free and encapsulated terminals in skin and near hairs; none of these has absolute specificity for a particular sensory modality. Special sensory organs are the olfactory, visual, acoustic, vestibular and taste receptors.

Proprioceptors respond to stimuli to deeper tissues, especially of the locomotor system, and are concerned with detecting movement, mechanical stresses and position. They include Golgi tendon organs, muscle spindles, Pacinian corpuscles, other endings in joints, and vestibular receptors. Proprioceptors are stimulated by the contraction of muscles, movements of joints and changes in the position of the body. They are essential for the coordination of muscles, the grading of muscular contraction, and the maintenance of equilibrium.

Interoceptors are found in the walls of the viscera, glands and vessels, where their terminations include free nerve endings, encapsulated terminals and endings associated with specialized epithelial cells. Nerve terminals are found in the layers of visceral walls and the adventitia of blood vessels, but the detailed structure and function of many of these endings are not well established. Encapsulated (lamellated) endings occur in the heart, adventitia and mesenteries. Free terminal arborizations occur in the endocardium, the endomysium of all muscles, and connective tissue generally. Tension produced by excessive muscular contraction or by visceral distension often causes pain, particularly in pathological states, which is frequently poorly localized and of a deep-seated nature. Visceral pain is often referred to the corresponding dermatome (see Fig. 16.10). Polymodal nociceptors (irritant receptors) respond to a variety of stimuli such as noxious chemicals or damaging mechanical stimuli. They are mainly the free endings of fine, unmyelinated fibres that are widely distributed in the epithelia of the alimentary and respiratory tracts; they may initiate protective reflexes.

They include: clear cells (so named because of their poor staining properties in routine preparations); argentaffin cells (reduce silver salts); argyrophil cells (absorb silver); small intensely fluorescent cells; peptide-producing cells (particularly of the hypothalamus, hypophysis, pineal and parathyroid glands, and placenta); Kulchitsky cells in the lungs; and paraneurones. Many cells of the dispersed (or diffuse) neuroendocrine system are derived embryologically from the neural crest. Some – in particular, cells from the gastrointestinal system – are now known to be endodermal in origin.

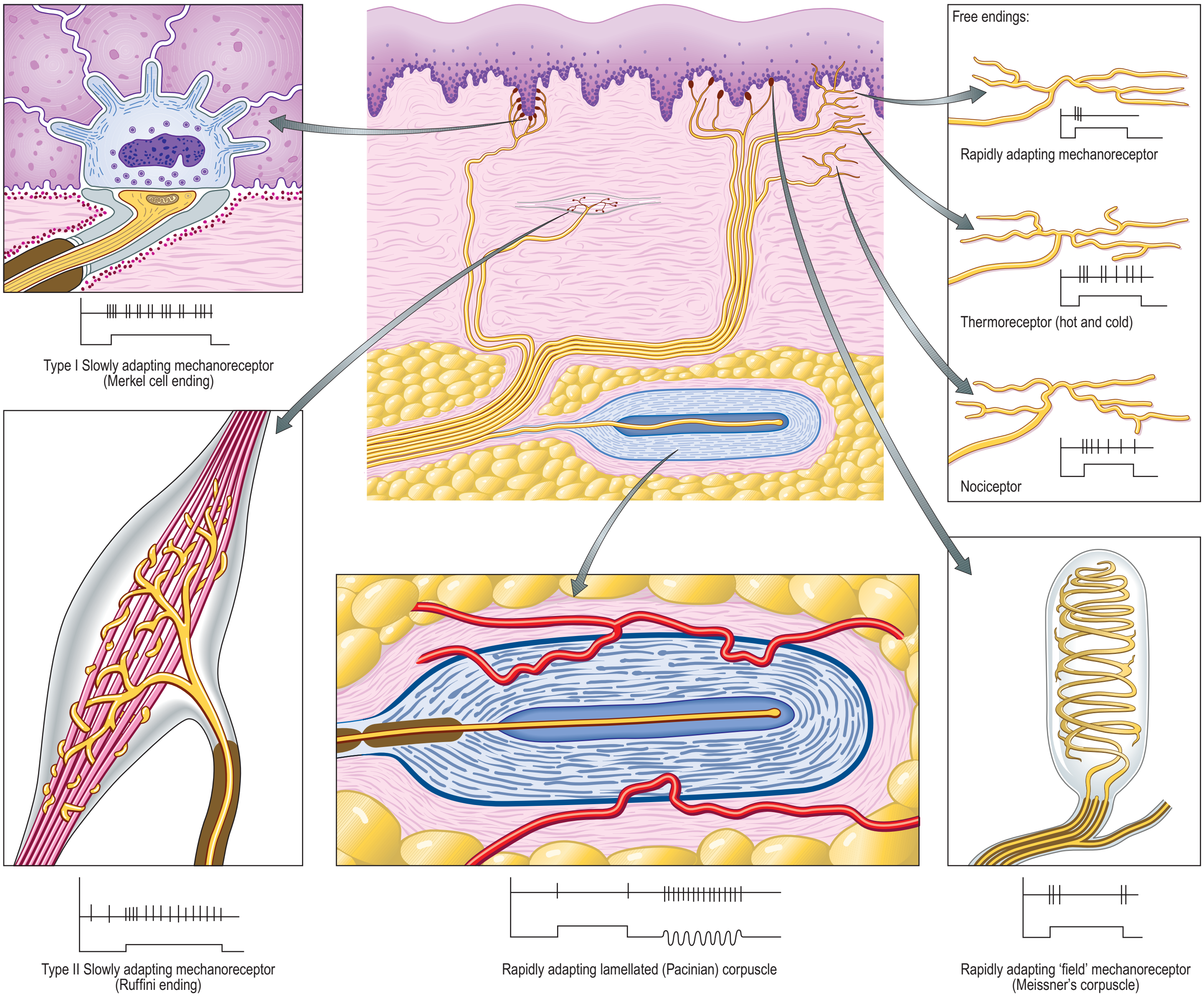

Fig. 3.27 Some major types of sensory ending of general afferent fibres (omitting neuromuscular, neurotendinous and hair-related types). The traces below each type of ending indicate (top) their response (firing rate (vertical lines) and adaption with time) to an appropriate stimulus (below) of the duration indicated. The Pacinian corpuscle's response to vibration (rapid sequence of on–off stimuli) is also shown.

Interoceptors include vascular chemoreceptors, e.g. the carotid body, and baroceptors, which are concerned with the regulation of blood flow and pressure and the control of respiration.

FREE NERVE ENDINGS

Sensory endings that branch to form plexuses occur in many sites (see Fig. 3.27). They occur in all connective tissues, including those of the dermis, fasciae, capsules of organs, ligaments, tendons, adventitia of blood vessels, meninges, articular capsules, periosteum, perichondrium, Haversian systems in bone, parietal peritoneum, walls of viscera and the endomysium of all types of muscle. They also innervate the epithelium of the skin, cornea, buccal cavity, and the alimentary and respiratory tracts and their associated glands. Within epithelia, free sensory endings lack Schwann cell ensheathment and are enveloped instead by epithelial cells. Afferent fibres from free terminals may be myelinated or unmyelinated but are always of small diameter and low conduction velocity. When afferent axons are myelinated, their terminal arborizations lack a myelin sheath. These terminals serve several sensory modalities. In the dermis, they may be responsive to moderate cold or heat (thermoreceptors); light mechanical touch (mechanoreceptors); damaging heat, cold or deformation (unimodal nociceptors); and damaging stimuli of several kinds (polymodal nociceptors). Similar fibres in deeper tissues may also signal extreme conditions, which are experienced, as with all nociceptors, as ache or pain. Free endings in the cornea, dentine and periosteum may be exclusively nociceptive.

Special types of free ending are associated with epidermal structures in the skin. They include terminals associated with hair follicles (peritrichial receptors), which branch from myelinated fibres in the deep dermal cutaneous plexus; the number, size and form of the endings are related to the size and type of hair follicle innervated. These endings respond mainly to movement when hair is deformed and belong to the rapidly adapting mechanoreceptor group.

Merkel tactile endings (see Commentary 1.3) lie either at the base of the epidermis or around the apical ends of some hair follicles, and most are innervated by large myelinated axons. Each axon expands into a disc that is applied closely to the base of a Merkel cell in the basal layer of the epidermis. The cells are believed to be derived from the epidermis, although a neural crest origin remains possible. They contain many large (50–100 nm) dense-core vesicles, presumably containing transmitters. Merkel endings are thought to be slow-adapting mechanoreceptors, responsive to sustained pressure and sensitive to the edges of applied objects. Their functions are controversial, however, and likely to be more varied.

ENCAPSULATED ENDINGS

Encapsulated endings are a major group of special endings that includes lamellated corpuscles of various kinds (e.g. Meissner's, Pacinian), Golgi tendon organs, neuromuscular spindles and Ruffini endings (see Fig. 3.27). They exhibit considerable variety in their size, shape and distribution but share a common feature: namely, that each axon terminal is encapsulated by non-excitable cells (Proske and Gandevia 2012).

Meissner's corpuscles

Meissner's corpuscles are found in the dermal papillae of all parts of the hand and foot, the anterior aspect of the forearm, the lips, palpebral conjunctiva and mucous membrane of the apical part of the tongue. They are most concentrated in thick hairless skin, especially of the finger pads, where there may be up to 24 corpuscles per cm^2 in young adults. Mature corpuscles are cylindrical in shape, approximately 80 µm long and 30 µm across, with their long axes perpendicular to the skin surface. Each corpuscle has a connective tissue capsule and central core composed of a stack of flat modified Schwann cells (Fig. 3.28). Meissner's corpuscles are rapidly adapting mechanoreceptors, sensitive to shape and textural changes in exploratory and discriminatory touch; their acute sensitivity provides the neural basis for reading Braille text.

Pacinian corpuscles

Pacinian corpuscles are situated subcutaneously in the palmar and plantar aspects of the hand and foot and their digits, the external genitalia, arm, neck, nipple, periosteal and interosseous membranes, and near joints and within the mesenteries (Fig. 3.29). They are oval, spherical or irregularly coiled and measure up to 2 mm in length and 100–500 µm or more across; the larger ones are visible to the naked eye. Each corpuscle has a capsule, an intermediate growth zone and a central core that contains an axon terminal. The capsule is formed by approximately 30 concentrically arranged lamellae of flat cells approximately 0.2 µm thick (see Fig. 3.28). Adjacent cells overlap and successive lamellae are separated by an amorphous proteoglycan matrix that contains circularly orientated collagen fibres, closely applied to the surfaces of the lamellar cells. The amount of collagen increases with age. The intermediate zone is cellular and its cells become incorporated into the capsule or core, so that it is not clearly defined in mature corpuscles. The core consists of approximately 60 bilateral, compacted lamellae lying on both sides of a central nerve terminal.

Each corpuscle is supplied by a myelinated axon, which initially loses its myelin sheath and subsequently loses its ensheathing Schwann cell at its junction with the core. The naked axon runs through the central axis of the core and ends in a slightly expanded bulb. It is in contact with the innermost core lamellae, is transversely oval and sends short projections of unknown function into clefts in the lamellae. The axon contains numerous large mitochondria, and minute vesicles, approximately 5 nm in diameter, which aggregate opposite the clefts. The cells of the capsule and core lamellae are thought to be specialized fibroblasts but some may be Schwann cells. Elastic fibrous tissue forms an overall external capsule to the corpuscle. Pacinian corpuscles are supplied by capillaries that accompany the axon as it enters the capsule.

Pacinian corpuscles act as very rapidly adapting mechanoreceptors. They respond only to sudden disturbances and are especially sensitive to very-high-frequency vibration. The rapidity may be partly due to the lamellated capsule acting as a high pass frequency filter, damping slow distortions by fluid movement between lamellar cells. Groups of corpuscles respond to pressure changes, e.g. on grasping or releasing an object.

Ruffini endings

Ruffini endings are slowly adapting mechanoreceptors. They are found in the dermis of thin, hairy skin, where they function as dermal stretch receptors and are responsive to maintained stresses in dermal collagen. They consist of the highly branched, unmyelinated endings of myelinated afferents. They ramify between bundles of collagen fibres within a spindle-shaped structure, which is enclosed partly by a fibrocellular sheath derived from the perineurium of the nerve. Ruffini endings appear electrophysiologically similar to Golgi tendon organs, which they resemble, although they are less organized structurally. Similar structures appear in joint capsules (see below).

Golgi tendon organs

Golgi tendon organs are found mainly near musculotendinous junctions (Fig. 3.30), where more than 50 may occur at any one site. Each terminal is closely related to a group of muscle fibres (up to 20) as they insert into the tendon. Golgi tendon organs are approximately 500 µm long and 100 µm in diameter, and consist of small bundles of tendon

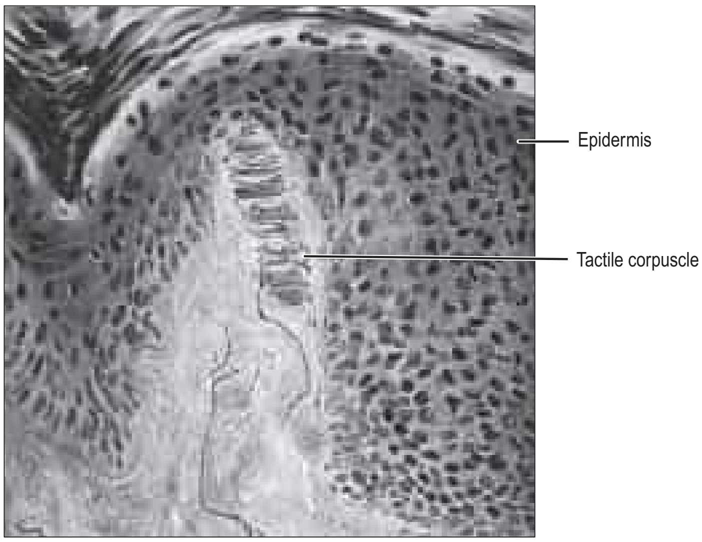

Fig. 3.28 A tactile Meissner's corpuscle in a dermal papilla in the skin, demonstrated using the modified Bielschowsky silver stain technique. (Courtesy of Professor N Cauna, University of Pittsburgh.)

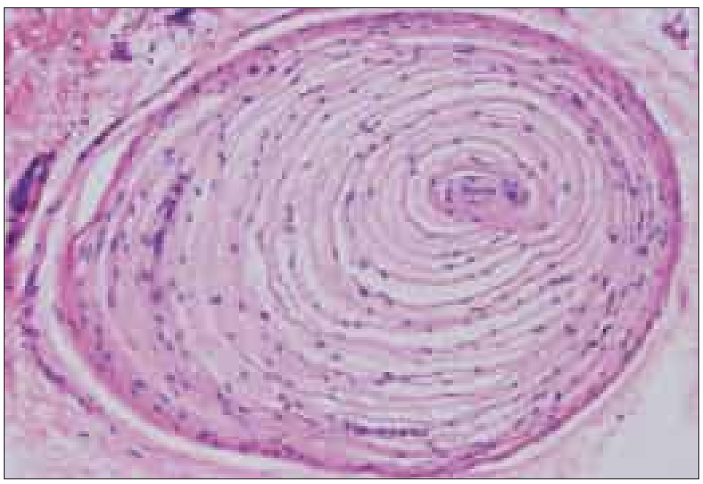

Fig. 3.29 A Pacinian corpuscle in human dermis. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

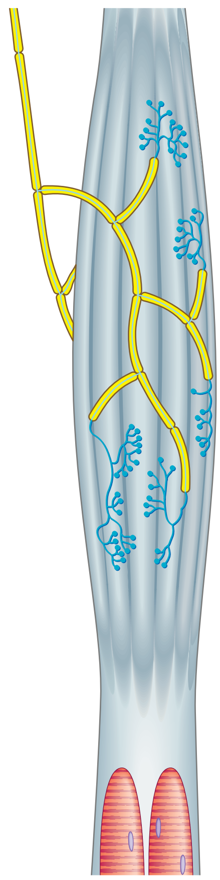

Fig. 3.30 The structure and innervation of a Golgi tendon organ. For clarity, the perineurium and endoneurium have been omitted to show the distribution of nerve fibres ramifying between the collagen fibre bundles of the tendon.

fibres enclosed in a delicate capsule. The collagen bundles (intrafusal fasciculi) are less compact than elsewhere in the tendon, the collagen fibres are smaller and the fibroblasts larger and more numerous. A single, thickly myelinated 1b afferent nerve fibre enters the capsule and divides. Its branches, which may lose their ensheathing Schwann cells, terminate in leaf-like enlargements containing vesicles and mitochondria, which wrap around the tendon. A basal lamina or process of Schwann cell cytoplasm separates the nerve terminals from the collagen bundles that constitute the tendon. Golgi tendon organs are activated by passive stretch of the tendon but are much more sensitive to active contraction of the muscle. They are important in providing proprioceptive information that complements the information coming from neuromuscular spindles. Their responses are slowly adapting and they signal maintained tension.

Neuromuscular spindles

Neuromuscular spindles are mechanosensors essential for proprioception (Boyd 1985). Each spindle contains a few small, specialized intrafusal muscle fibres, innervated by both sensory and motor nerve fibres (**Figs 3.31**–**3.32**). The whole is surrounded equatorially by a fusiform spindle capsule of connective tissue, consisting of an outer perineurium-like sheath of flattened fibroblasts and collagen, and an inner sheath that forms delicate tubes around individual intrafusal fibres (**Fig. 3.33**). A gelatinous fluid rich in glycosaminoglycans fills the space between the two sheaths.

There are usually 5–14 intrafusal fibres (the number varies between muscles) and two major types of fibre, nuclear bag and nuclear chain fibres, which are distinguished by the arrangement of nuclei in their sarcoplasm. In nuclear bag fibres, an equatorial cluster of nuclei makes the fibre bulge slightly, whereas the nuclei in nuclear chain fibres form a single axial row. Nuclear bag fibres are subdivided into bag1 and bag2 fibres, are greater in diameter than chain fibres and extend beyond the surrounding capsule to the endomysium of nearby extrafusal muscle fibres. Nuclear chain fibres are attached at their poles to the capsule or to the sheaths of nuclear bag fibres.

The intrafusal fibres resemble typical skeletal muscle fibres, except that the zone of myofibrils is thin around the nuclei. Dynamic bag1 fibres generally lack M lines, possess little sarcoplasmic reticulum, and have an abundance of mitochondria and oxidative enzymes, but little glycogen. Static bag2 fibres have distinct M lines and abundant glycogen. Nuclear chain fibres have marked M lines, sarcoplasmic reticulum and T-tubules, and abundant glycogen, but few mitochondria. Each fibre type carries distinct myosin heavy chain isoforms. These variations reflect the contractile properties of different intrafusal fibres.

Muscle spindles receive two types of sensory innervation via the unmyelinated terminations of large myelinated axons. Primary (anulospiral) endings are equatorially placed and form spirals around the nucleated parts of intrafusal fibres. They are the endings of large sensory fibres (group Ia afferents), each of which sends branches to a number of intrafusal muscle fibres. Each terminal lies in a deep sarcolemmal groove in the spindle plasma membrane beneath its basal lamina. Secondary (flower spray) endings, which may be spray-shaped or anular, are largely confined to bag2 and nuclear chain fibres, and are the branched terminals of somewhat thinner myelinated (group II) afferents. They are varicose and spread in a narrow band on both sides of the primary endings. They lie close to the sarcolemma, though not in grooves. In essence, primary endings are rapidly adapting, while secondary endings have a regular, slowly adapting response to static stretch.

There are three types of motor endings in muscle spindles. Two are from fine, myelinated, fusimotor (γ) efferents and one is from myelinated (β) efferent collaterals of axons that supply extrafusal slow twitch muscle fibres. The fusimotor efferents terminate nearer the equatorial region, where their terminals either resemble the motor end-plates of extrafusal fibres (plate endings) or are more diffuse (trail endings). Stimulation of the fusimotor and β-efferents causes contraction of the intrafusal fibres and, consequently, activation of their sensory endings.

Muscle spindles signal the length of extrafusal muscle both at rest and throughout contraction and relaxation, the velocity of their contraction and changes in velocity. These modalities may be related to the different behaviours of the three major types of intrafusal fibre and their sensory terminals. The sensory fusimotor endings of one type of nuclear bag fibre (dynamic bag1) are particularly concerned with signalling rapid changes in length that occur during movement, whilst those of the second bag fibre type (static bag2) and of chain fibres are less responsive to movement. These elements can therefore detect complex changes in the state of the extrafusal muscle surrounding spindles and can signal fluctuations in length, tension, velocity of length change and

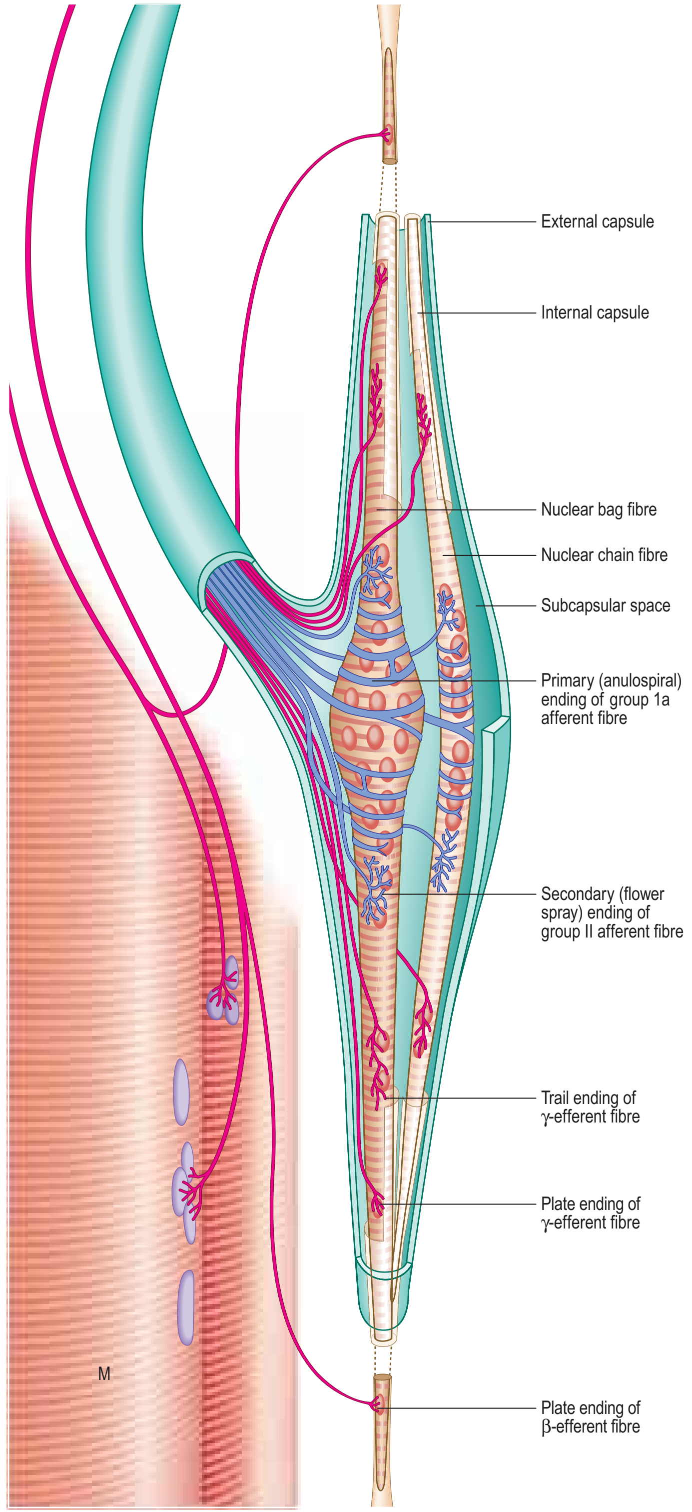

Fig. 3.31 A neuromuscular spindle, showing nuclear bag and nuclear chain fibres within the spindle capsule (green); these are innervated by the sensory anulospiral and 'flower spray' afferent fibre endings (blue) and by the γ and β fusimotor (efferent fibre) endings (red). The β efferent fibres are collaterals of fibres innervating extrafusal slow twitch muscle cells (M).

acceleration. Moreover, they are under complex central control; efferent (fusimotor) nerve fibres, by regulating the strength of contraction, can adjust the length of the intrafusal fibres and thereby the responsiveness of spindle sensory endings. In summary, the organization of spindles allows them to monitor muscle conditions actively in order to compare intended and actual movements, and to provide a detailed input to spinal, cerebellar, extrapyramidal and cortical centres about the state of the locomotor apparatus.

JOINT RECEPTORS

The arrays of receptors situated in and near articular capsules provide information on the position, movements and stresses acting on joints.

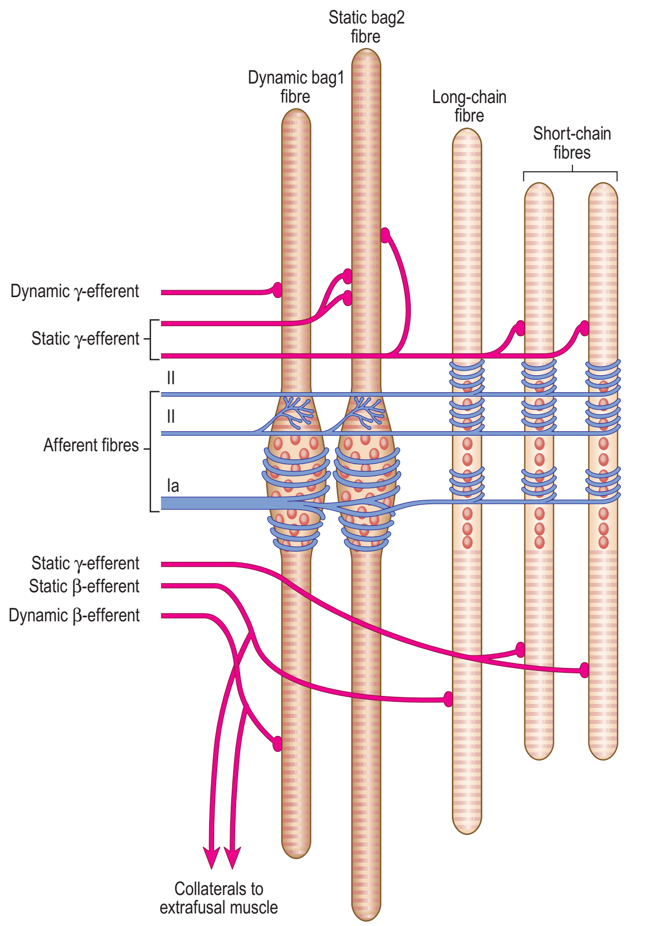

Fig. 3.32 Nuclear bag and nuclear chain fibres in a neuromuscular spindle. Dynamic β- and γ-efferents innervate dynamic bag1 intrafusal fibres, whereas static β- and γ-efferents innervate static bag2 and nuclear chain intrafusal fibres.

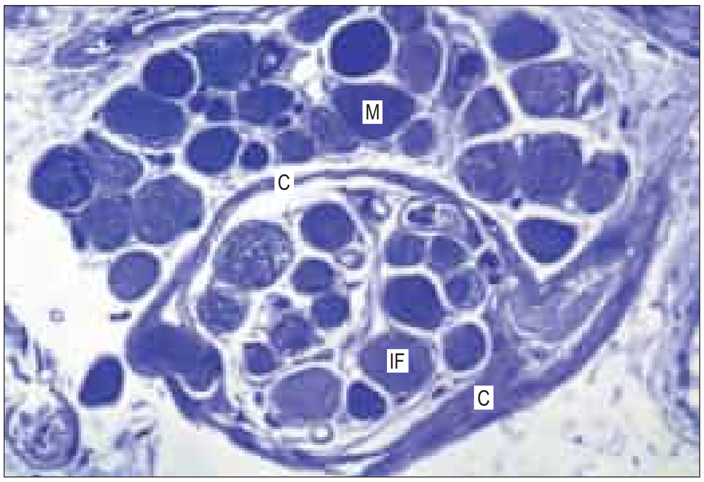

Fig. 3.33 A neuromuscular spindle in transverse section in a human extraocular muscle. The spindle capsule (C) encloses intrafusal fibres (IF) of varying diameters. Typical muscle fibres (M) in transverse section are shown above the spindle. Toluidine blue stained resin section.

Structural and functional studies have demonstrated at least four types of joint receptor; their proportions and distribution vary with site. Three are encapsulated endings, the fourth a free terminal arborization.

Type I endings are encapsulated corpuscles of the slowly adapting mechanoreceptor type and resemble Ruffini endings. They lie in the superficial layers of the fibrous capsules of joints in small clusters and are innervated by myelinated afferent axons. Being slowly adapting, they provide awareness of joint position and movement, and respond to patterns of stress in articular capsules. They are particularly common in joints where static positional sense is necessary for the control of posture (e.g. hip, knee).

Type II endings are lamellated receptors and resemble small versions of the large Pacinian corpuscles found in general connective tissue. They occur in small groups throughout joint capsules, particularly in the deeper layers and other articular structures (e.g. the fat pad of the temporomandibular joint). They are rapidly adapting, low-threshold mechanoreceptors, sensitive to movement and pressure changes, and they respond to joint movement and transient stresses in the joint capsule. They are supplied by myelinated afferent axons but are probably not involved in the conscious awareness of joint sensation.

Type III endings are identical to Golgi tendon organs in structure and function; they occur in articular ligaments but not in joint capsules. They are high-threshold, slowly adapting receptors and may serve, at least in part, to prevent excessive stresses at joints by reflex inhibition of the adjacent muscles. They are innervated by large myelinated afferent axons.

Type IV endings are free terminals of myelinated and unmyelinated axons that ramify in articular capsules and the adjacent fat pads, and around the blood vessels of the synovial layer. They are high-threshold, slowly adapting receptors and are thought to respond to excessive movements, providing a basis for articular pain.

NEUROMUSCULAR JUNCTIONS

SKELETAL MUSCLE

The most intensively studied effector endings are those that innervate muscle, particularly skeletal muscle. All neuromuscular (myoneural) junctions are axon terminals of motor neurones. They are specialized for the release of neurotransmitter on to the sarcolemma of skeletal muscle fibres, causing a change in their electrical state that leads to contraction. Each axon branches near its terminal to innervate from several to hundreds of muscle fibres, the number depending on the precision of motor control required (Shi et al 2012).

The detailed structure of a motor terminal varies with the type of muscle innervated. Two major types of ending are recognized, innervating either extrafusal muscle fibres or the intrafusal fibres of neuromuscular spindles. In the former type, each axonal terminal usually ends midway along a muscle fibre in a discoidal motor end-plate (**Fig. 3.34A**), and usually initiates action potentials that are rapidly conducted to all parts of the muscle fibre. In the latter type, the axon gives off numerous branches that form a cluster of small expansions extending along the muscle fibre; in the absence of propagated muscle excitation, these excite the fibre at several points. Both types of ending are associated with a specialized receptive region of the muscle fibre, the sole plate, where a number of muscle cell nuclei are grouped within the granular sarcoplasm.

The sole plate contains numerous mitochondria, endoplasmic reticulum and Golgi complexes. The terminal branches of the axon are plugged into shallow grooves in the surface of the sole plate (primary clefts), from where numerous pleats extend for a short distance into the underlying sarcoplasm (secondary clefts) (**Fig. 3.34B,C**). The axon terminal contains mitochondria and many clear, 60 nm spherical vesicles similar to those in presynaptic boutons, which are clustered over the zone of membrane apposition. It is ensheathed by Schwann cells whose cytoplasmic projections extend into the synaptic cleft. The plasma membranes of the axon terminal and the muscle cell are separated by a 30–50 nm gap and an interposed basal lamina, which follows the surface folding of the sole-plate membrane into the secondary clefts. Endings of fast and slow twitch muscle fibres differ in detail: the sarcolemmal grooves are deeper, and the presynaptic vesicles more numerous, in the fast fibres.

Junctions with skeletal muscle are cholinergic: the release of ACh changes the ionic permeability of the muscle fibre (Sine 2012). Clustering of ACh receptors at the neuromuscular junction depends in part on the presence in the muscle basal lamina of agrin, which is secreted by the motor neurone, and is important in establishing the postjunctional molecular machinery. When the depolarization of the sarcolemma reaches a particular threshold, it initiates an action potential in the sarcolemma, which is then propagated rapidly over the whole cell surface and also deep within the fibre via the invaginations (T-tubules) of the sarcolemma, causing contraction. The amount of ACh released by the arrival of a single nerve impulse is sufficient to trigger an action potential. However, because ACh is very rapidly hydrolysed by the enzyme AChE, present at the sarcolemmal surface of the sole plate, a single nerve impulse only gives rise to one muscle action potential, i.e. there is a one-to-one relationship between neuronal and muscle action potentials. Thus the contraction of a muscle fibre is controlled by the firing frequency of its motor neurone.

Neuromuscular junctions are partially blocked by high concentrations of lactic acid, as in some types of muscle fatigue.

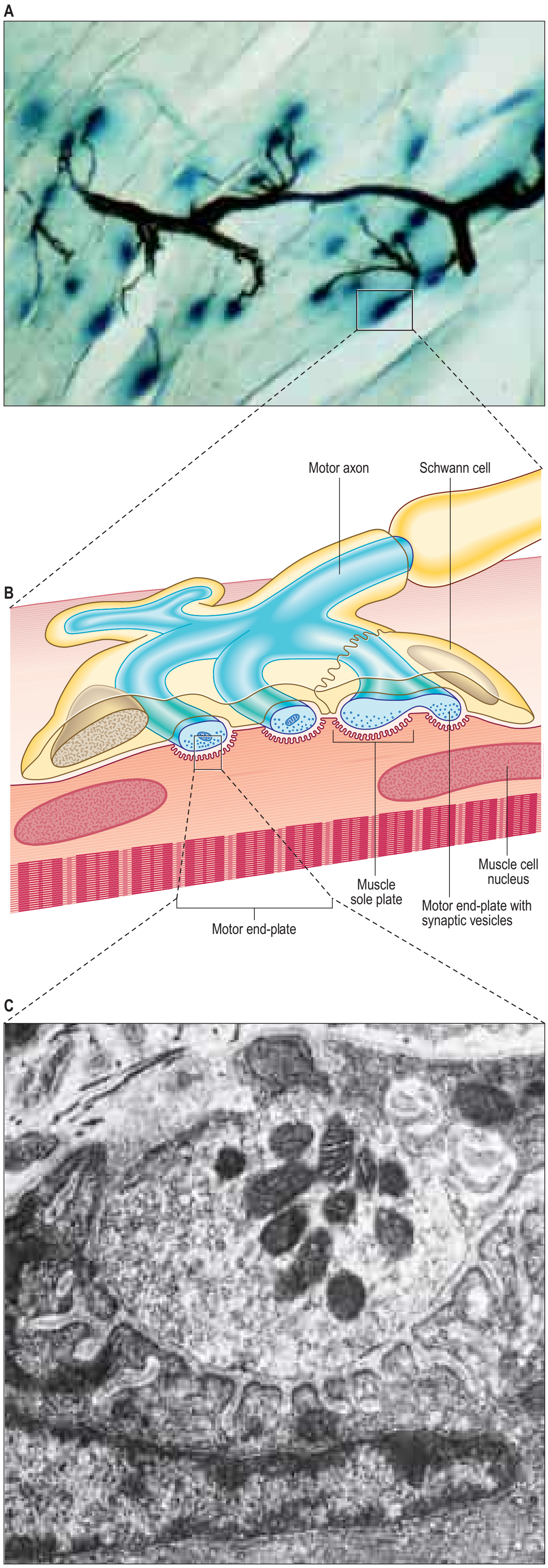

Fig. 3.34 The neuromuscular junction. **A**, Whole-mount preparation of teased skeletal muscle fibres (pale, faintly striated, diagonally orientated structures). The terminal part of the axon (silver-stained, brown) branches to form motor end-plates on adjacent muscle fibres. The sole plate recesses in the sarcolemma, into which the motor end-plates fit, are demonstrated by the presence of acetylcholinesterase (shown by enzyme histochemistry, blue). **B**, The axonal motor end-plate and the deeply infolded sarcolemma. **C**, Electron micrograph showing the expanded motor end-plate of an axon filled with vesicles containing synaptic transmitter (ACh) (above); the deep infoldings of the sarcolemmal sole plate (below) form subsynaptic gutters. (A, Courtesy of Dr Norman Gregson, Division of Neurology, GKT School of Medicine, London. C, Courtesy of Professor DN Landon, Institute of Neurology, University College London.)

AUTONOMIC MOTOR TERMINATIONS

Autonomic neuromuscular junctions differ in several important ways from the skeletal neuromuscular junction and from synapses in the CNS and PNS. There is no fixed junction with well-defined pre- and postjunctional specializations. Unmyelinated, highly branched, postganglionic autonomic axons become beaded or varicose as they reach the effector smooth muscle. These varicosities are not static but move along axons. They are packed with mitochondria and vesicles containing neurotransmitters, which are released from the varicosities during conduction of an impulse along the axon. The distance (cleft) between the varicosity and smooth muscle membrane varies considerably depending on the tissue, from 20 nm in densely innervated structures such as the vas deferens to 1–2 µm in large elastic arteries. Unlike skeletal muscle, the effector tissue is a muscle bundle rather than a single cell. Gap junctions between individual smooth muscle cells are low-resistance pathways that allow electronic coupling and the spread of activity within the effector bundle; they vary in size from punctate junctions to junctional areas of more than 1 µm in diameter.

Adrenergic sympathetic postganglionic terminals contain dense-cored vesicles. Cholinergic terminals, which are typical of all parasympathetic and some sympathetic endings, contain clear spherical vesicles like those in the motor end-plates of skeletal muscle. A third category of autonomic neurones has non-adrenergic, non-cholinergic endings that contain a wide variety of chemicals with transmitter properties. ATP is a neurotransmitter at these terminals, which express purinergic receptors (Burnstock et al 2011). The axons typically contain large, 80–200 nm, dense opaque vesicles, congregated in varicosities at intervals along their length. These terminals are formed in many sites, including the lungs, blood vessel walls, the urogenital tract and the external muscle layers and sphincters of the gastrointestinal tract. In the intestinal wall, neuronal somata lie in the myenteric plexus, and their axons spread caudally for a few millimetres, mainly to innervate circular muscle. Purinergic neurones are under cholinergic control from preganglionic sympathetic neurones. Their endings mainly hyperpolarize smooth muscle cells, causing relaxation, e.g. preceding peristaltic waves, opening sphincters and, probably, causing reflex distension in gastric filling. Autonomic efferents innervate exocrine glands, myoepithelial cells, adipose tissue (noradrenaline (norepinephrine) released from postganglionic sympathetic axons binds to β_3-receptors on adipocytes to stimulate lipolysis) and the vasculature and parenchymal fields of lymphocytes and associated cells in several lymphoid organs, including the thymus, spleen and lymph nodes.

CNS–PNS TRANSITION ZONE

The transition between CNS and PNS usually occurs some distance from the point at which nerve roots emerge from the brain or the spinal cord. The segment of root that contains components of both CNS and PNS tissue is called the CNS–PNS transition zone (TZ). All axons in the PNS, other than postganglionic autonomic neurones, cross such a TZ. Macroscopically, as a nerve root is traced towards the spinal cord or the brain, it splits into several thinner rootlets that may, in turn, subdivide into minirootlets. The TZ is located within either rootlet or minirootlet (Fig. 3.35). The arrangement of roots and rootlets varies according to whether the root trunk is ventral, dorsal or cranial. Thus, in dorsal roots, the main root trunk separates into a fan of rootlets and minirootlets that enter the spinal cord in sequence along the dorsolateral sulcus. In certain cranial nerves, the minirootlets come together central to the TZ and enter the brain as a stump of white matter.

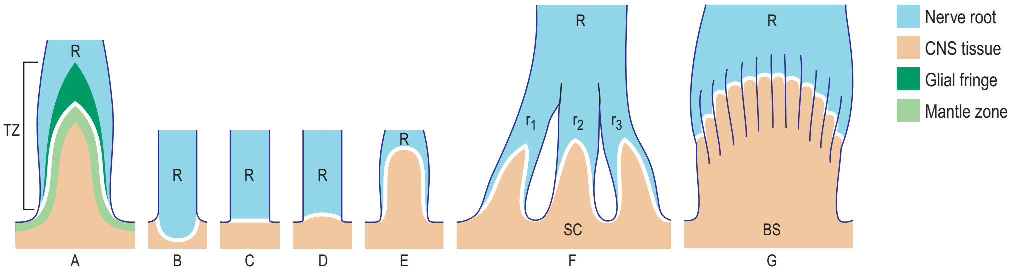

Fig. 3.35 The nerve root–spinal cord junction. **A**–**G**, Different CNS–PNS borderline arrangements. **A**, A pointed borderline. The extent of the transitional zone (TZ) is indicated. **B**–**G**, Glial fringe omitted. **B**, A concave borderline (white line) and inverted TZ. **C**, A flat borderline situated at the level of the root (R)–spinal cord junction. **D** and **E**, A convex, dome-shaped borderline; the CNS expansion into the rootlet is moderate in D and extensive in E. **F**, The root (R) splits into rootlets (r), each with its own TZ and attaching separately to the spinal cord (SC). **G**, The arrangement found in several cranial nerve roots (e.g. vestibulocochlear nerve). The PNS component of the root separates into a bundle of closely packed minirootlets, each equipped with a TZ. The minirootlets reunite centrally. BS, brainstem. (Adapted with permission from Dyck PJ, Thomas PK, Griffin JW, et al (eds) Peripheral Neuropathy, 3rd ed. Philadelphia: Saunders, 1993.)

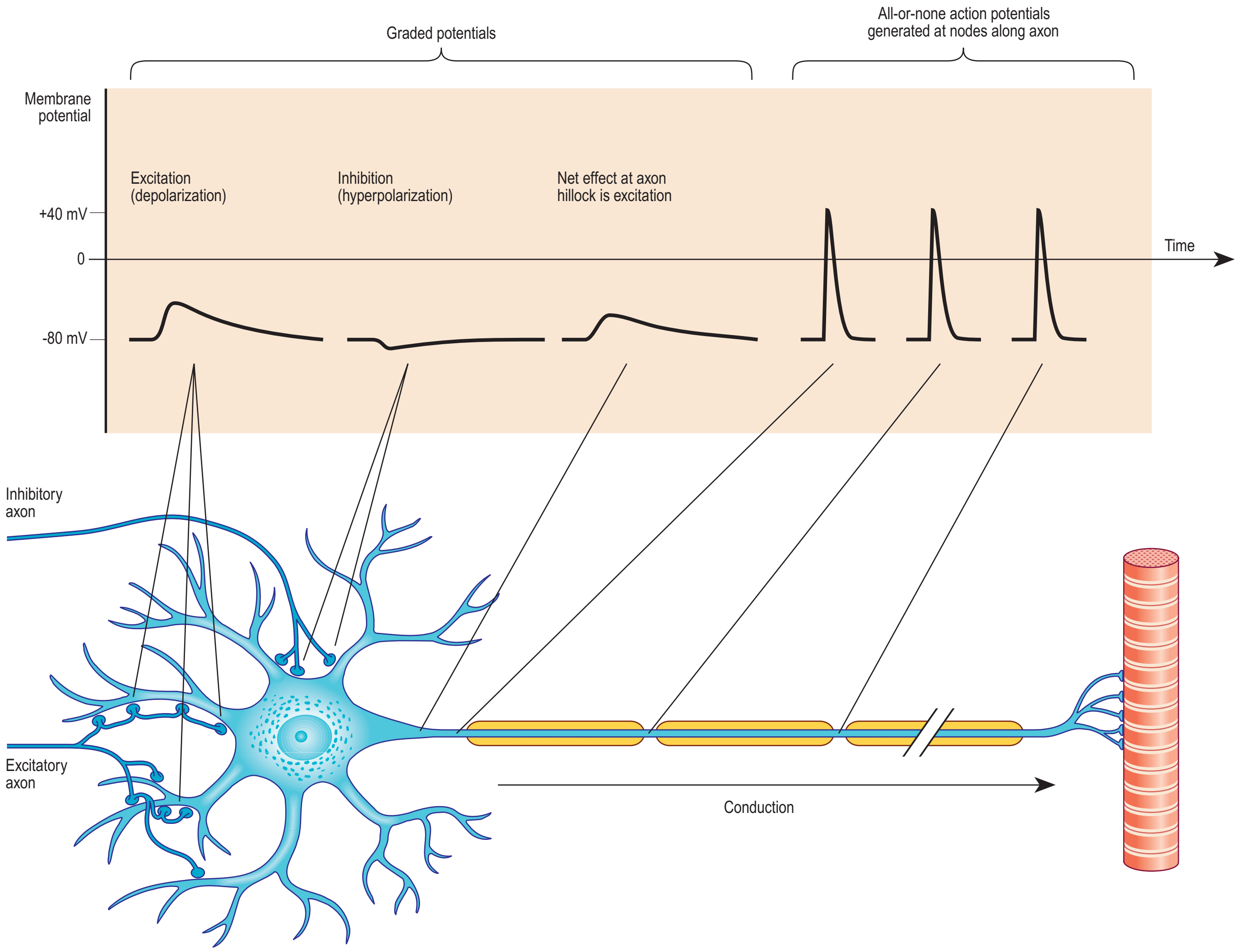

Fig. 3.36 The types of change in electrical potential that can be recorded across the cell membrane of a motor neurone at the points indicated. Excitatory and inhibitory synapses on the surfaces of the dendrites and soma cause local graded changes of potential that summate at the axon hillock and may initiate a series of all-or-none action potentials, which in turn are conducted along the axon to the effector terminals.

Microscopically, the TZ is characterized by an axial CNS compartment surrounded by a PNS compartment. The zone lies more peripherally in sensory nerves than in motor nerves, but in both, the apex of the TZ is described as a glial dome, whose convex surface is usually directed distally. The centre of the dome consists of fibres with a typical CNS organization, surrounded by an outer mantle of astrocytes (corresponding to the glia limitans). From this mantle, numerous glial processes project into the endoneurial compartment of the peripheral nerve, where they interdigitate with its Schwann cells. The astrocytes form a loose reticulum through which axons pass. Peripheral myelinated axons usually cross the zone at a node of Ranvier, which is here termed a PNS–CNS compound node.

Boundary cap (BC) cells are neural crest derivatives that form transient, discrete clusters localized at the presumptive dorsal root entry zones and motor exit points of the embryonic spinal cord. They are thought to prevent cell mixing at these interfaces not only by helping dorsal root ganglion afferents navigate their path to targets in the spinal cord but also by inhibiting motoneurone cell bodies exiting to the periphery. For further reading, see Zujovic et al (2011).

CONDUCTION OF THE NERVOUS IMPULSE

All cells generate a steady electrical potential across their plasma membrane (the membrane potential). This potential is generated by an uneven distribution of potassium ions across the membrane (higher in the intracellular compartment than in the extracellular compartment), and by a selective permeability of the membrane for potassium (Fig. 3.36). The distribution of sodium ions is opposite to that for potassium ions, but at rest the sodium conductance of the membrane is low. In neurones this membrane potential is known as the resting potential, and amounts to approximately −60 mV (potential inside the cell measured relative to the outside of the cell). Non-excitable cells have an even higher membrane potential. Neurones receive, conduct and transmit information by changes in membrane conductance for sodium, potassium, calcium or chloride ions. Increase in the sodium or calcium conductance causes an influx of these ions and results in a depolarization of the cell, while chloride influx or potassium efflux results in hyperpolarization. Plasma membrane permeability to these ions is altered by the opening or closing of ion-specific transmembrane channels, triggered by voltage changes or chemical signals such as transmitters (Catterall 2010).

Chemically triggered ionic fluxes may be either direct, where the chemical agent (neurotransmitter) binds to the channel itself to cause it to open, or indirect, where the neurotransmitter is bound by a transmembrane receptor molecule that activates a complex second messenger system within the cell to open separate transmembrane channels. Electrically induced changes in membrane potential depend on the presence of voltage-sensitive ion channels, which, when the transmembrane potential reaches a critical level, open to allow the influx or efflux of specific ions. In all cases, the channels remain open only transiently, and the numbers that open and close determine the total flux of ions across the membrane (Bezanilla 2008).

The types and concentrations of transmembrane channels and related proteins, and therefore the electrical activity of the membranes, vary in different parts of the cell. Dendrites and neuronal somata depend mainly on neurotransmitter action and show graded potentials, whereas axons have voltage-gated channels that give rise to action potentials.

In graded potentials, a flow of current occurs when a synapse is activated; the influence of an individual synapse on the membrane potential of neighbouring regions decreases with distance. Thus synapses on the distal tips of dendrites may, on their own, have relatively little effect on the membrane potential of the cell body. The electrical state of a neurone therefore depends on many factors, including the numbers and positions of thousands of excitatory and inhibitory synapses, their degree of activation, and the branching pattern of the dendritic tree and geometry of the cell body. The integrated activity directed towards the neuronal cell body is converted to an output directed away from the soma at the site where the axon leaves the cell body, at its junction with the axon hillock. Voltage-sensitive channels are concentrated at this trigger zone, the axon initial segment, and when this region is sufficiently depolarized, an action potential is generated and is subsequently conducted along the axon.

ACTION POTENTIAL

The action potential is a brief, self-propagating reversal of membrane polarity. It depends on an initial influx of sodium ions, which causes a reversal of polarity to about +20 mV, followed by a rapid return towards the resting potential as potassium ions flow out. The rapid reversal process is completed in approximately 0.5 msec, followed by a slower recovery phase of up to 5 msec, when the resting potential is even hyperpolarized. Once the axon hillock reaches threshold, propagation of the action potential is independent of the initiating stimulus; thus the size and duration of action potentials are always the same (described as all-or-none) for a particular neurone, no matter how much a stimulus may exceed the threshold value.

Once initiated, an action potential spreads spontaneously and at a relatively constant velocity, within the range of 4–120 m/s. Conduction velocity depends on a number of factors related to the way in which the current spreads, e.g. axonal cross-sectional area, the numbers and positioning of ion channels, and membrane capacitance (influenced particularly by the presence of myelin). In axons lacking myelin, action potential conduction is analogous to a flame moving along a fuse. Just as each segment of fuse is ignited by its upstream neighbour, each segment of axon membrane is driven to threshold by the depolarization of neighbouring membrane. Sodium channels within the newly depolarized segment open and positively charged sodium ions enter, driving the local potential inside the axon towards positive values. This inward current in turn depolarizes the neighbouring, downstream, non-depolarized membrane, and the cyclic propagation of the action potential is completed. Several milliseconds after the action potential, the sodium channels are inactivated, a period known as the refractory period. The length of the refractory period determines the maximum frequency at which action potentials can be conducted along a nerve fibre; it varies in different neurones and affects the amount of information that can be carried by an individual fibre.

Myelinated fibres are electrically insulated by their myelin sheaths along most of their lengths, except at nodes of Ranvier. The distance between nodes, referred to as the internodal distance, is directly related to axon diameter and varies between 0.2 and 2.0 mm. Voltage-gated sodium channels are clustered at nodes, and the nodal membrane is the only place where high densities of inward sodium current can be generated across the axon membrane. Conduction in myelinated axons is self-propagating, but instead of physically adjacent regions of membrane acting to excite one another (as occurs along unmyelinated axons), it is the depolarization occurring in the neighbouring upstream node that excites a node to threshold. Reaching threshold causes the sodium channels at the node to open and generate inward sodium current, but instead of this acting on the adjacent membrane, the high resistance and low capacitance of the myelin sheath directs the current towards the next downstream node, exciting it to threshold and completing the cycle. The action potential thus jumps from node to node, a process known as saltatory conduction, which greatly increases the conduction velocity.

A number of disorders of the CNS and PNS include demyelination as a characteristic feature. Perhaps most common amongst these is multiple sclerosis, which is characterized by primary demyelination at scattered sites within the CNS (it is now recognized that axonal loss also contributes to the progression of multiple sclerosis). Primary demyelination is the loss of the myelin sheath with axonal preservation, and is usually segmental, i.e. it rarely extends along the entire length of an affected axon. The phenomenon is associated with conduction block because the newly exposed, previously internodal, axolemma contains relatively few voltage-sensitive Na^+ channels. There is experimental evidence that conduction can be restored in some demyelinated axons, and experimental and clinical evidence that remyelinated axons can conduct at near-normal speeds, because even though their sheaths are thinner than the original myelin sheaths, the safety factor (i.e. the factor by which the outward current at a quiescent node next to an excited node exceeds the minimum current required to evoke a response) is greater than 1. The myelin loss that occurs in the early stages of Wallerian degeneration in both CNS and PNS, usually distal to a site of trauma but also in response to a prolonged period of ischaemia or exposure to a neuronotoxic substance, is accompanied by axonal degeneration (the term secondary demyelination is sometimes used to describe this form of myelin loss).

Axonal conduction is naturally unidirectional, from dendrites and soma to axon terminals. When an action potential reaches the axonal terminals, it causes depolarization of the presynaptic membrane, and as a result, quanta of neurotransmitter (which correspond to the content of individual vesicles) are released to change the degree of excitation of the next neurone, muscle fibre or glandular cell.

Bonus e-book image

Fig. 3.1 A section through the human cerebellum stained to show the arrangement in the brain of the central white matter and the highly folded outer grey matter.

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